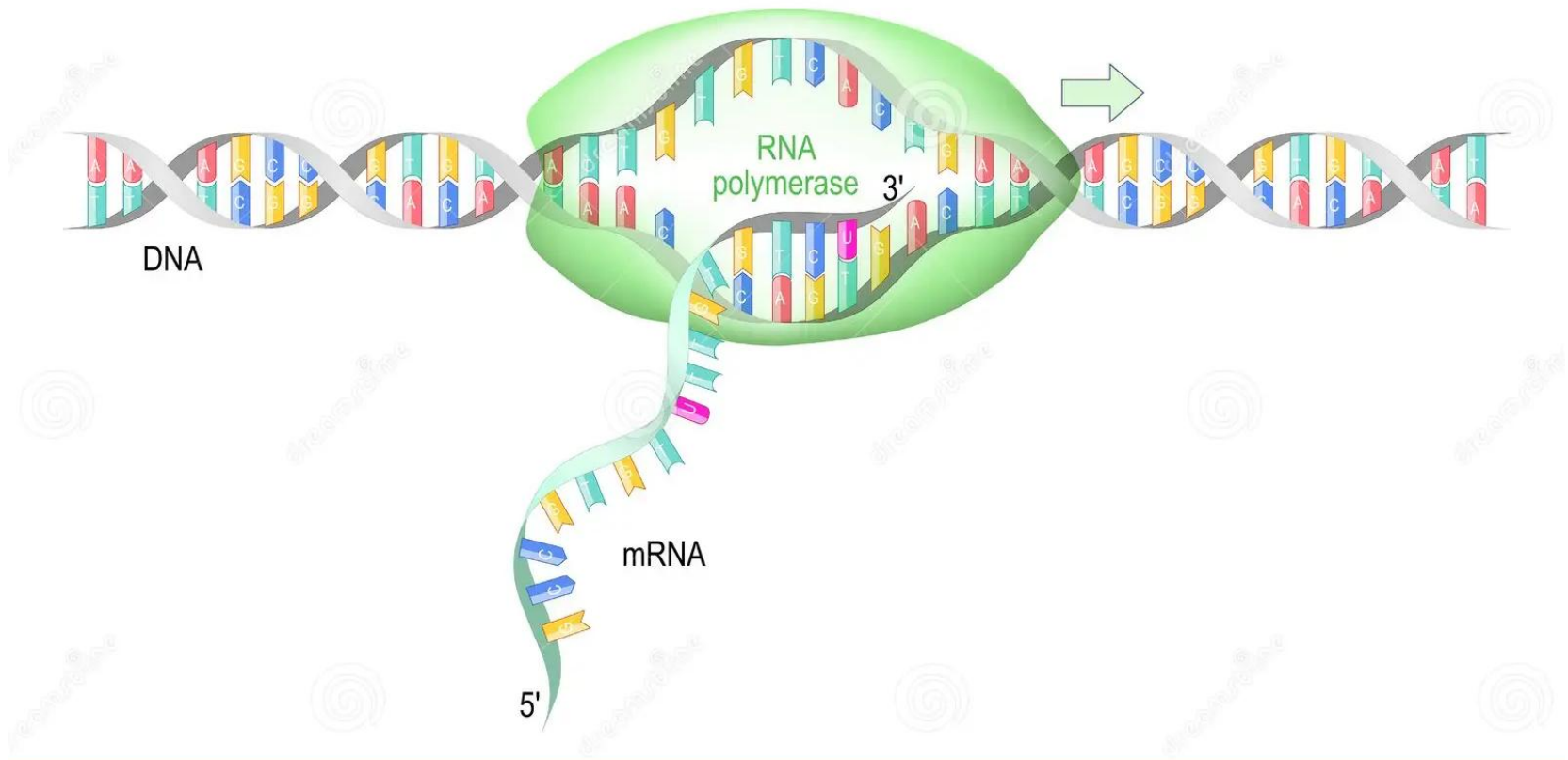
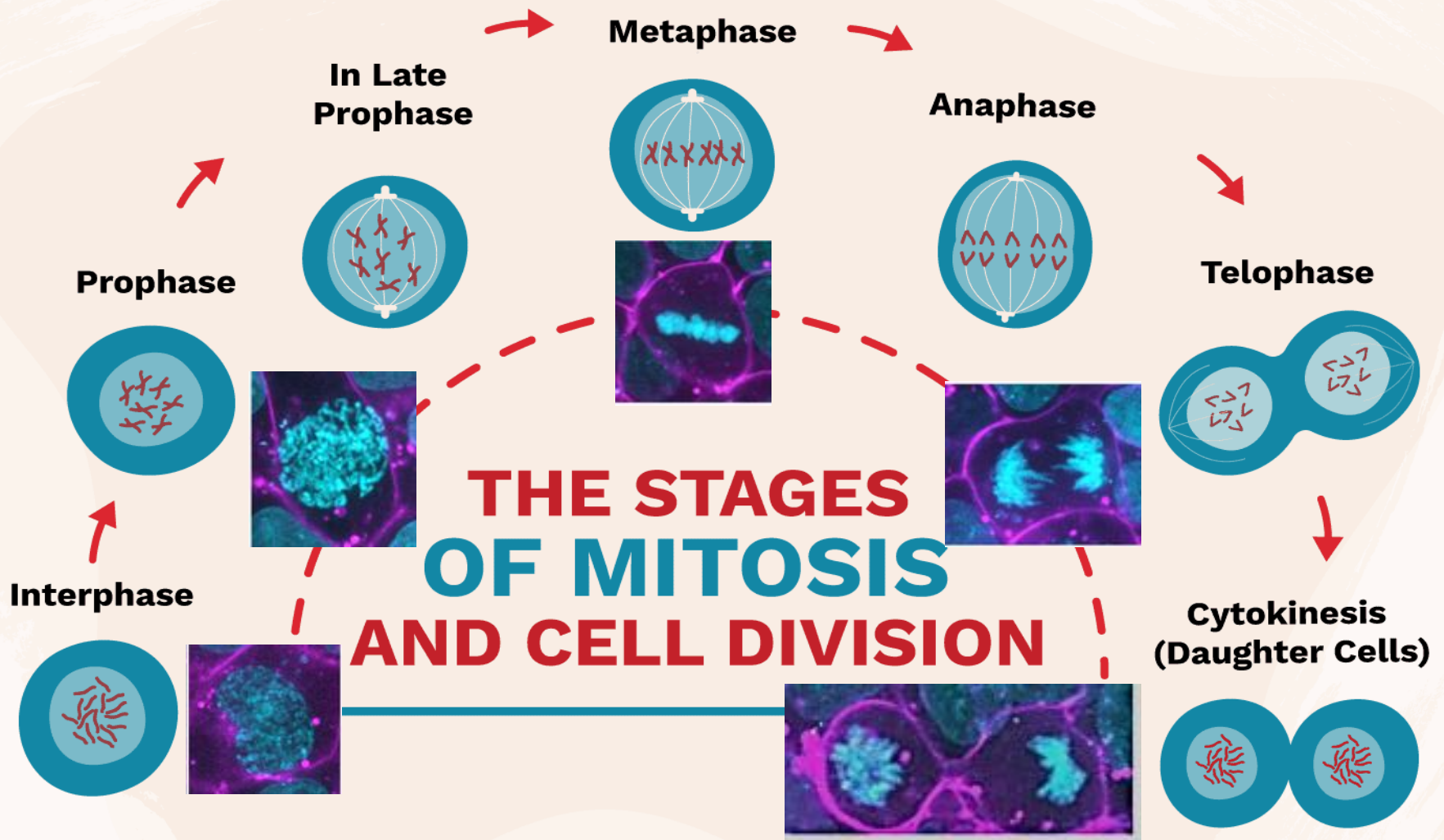


The code of life

La Trascrizione (the process of copying a segment of DNA into RNA)

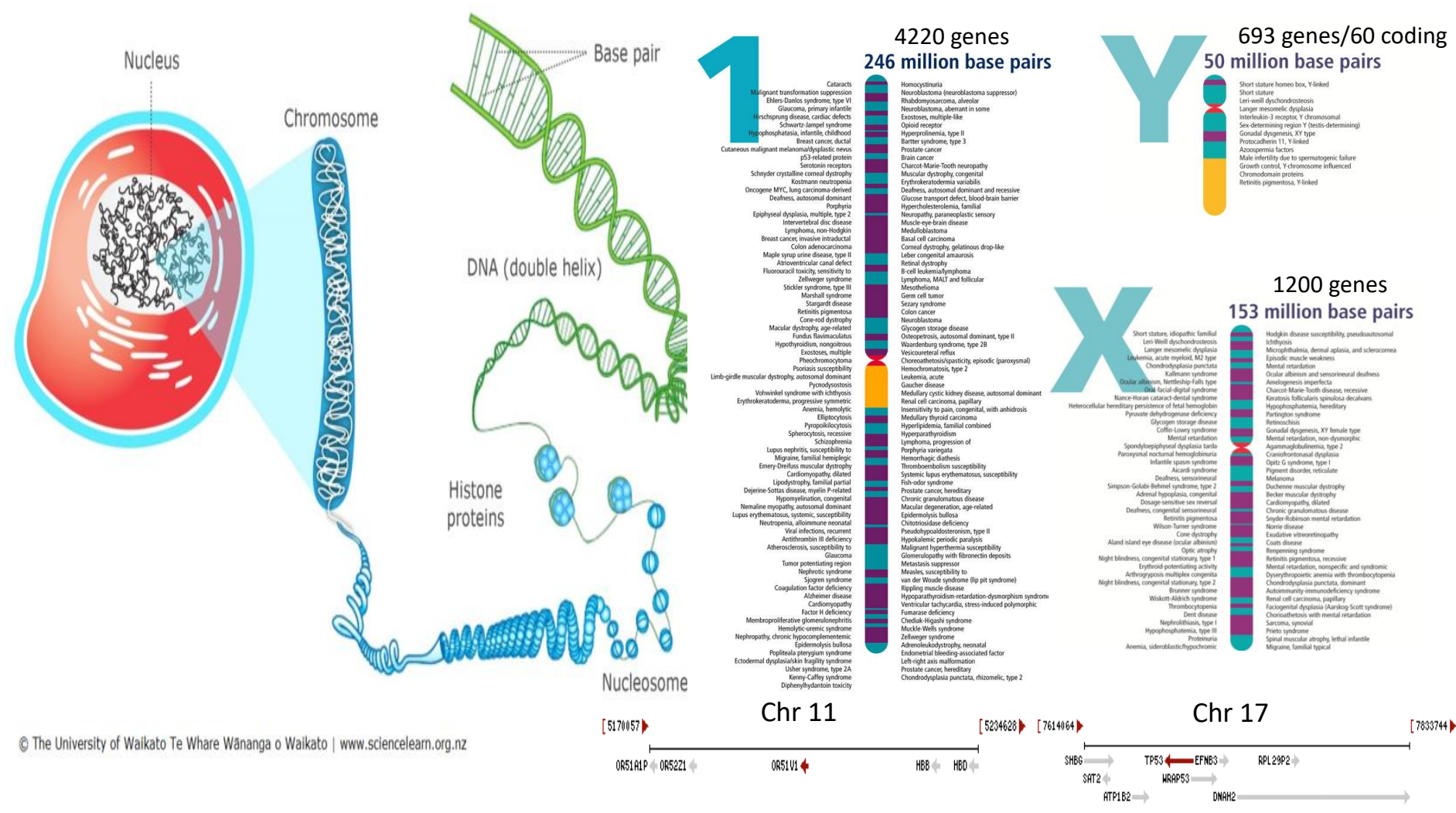


The code of life



The Code of Life: i cromosomi

All'interno di ciascun cromosoma sequenze definite di nucleotide costituiscono i geni



The Code of Life: Geni ed RNA

Durante la TRASCRIZIONE il gene viene usato come stampo per la sintesi di RNA

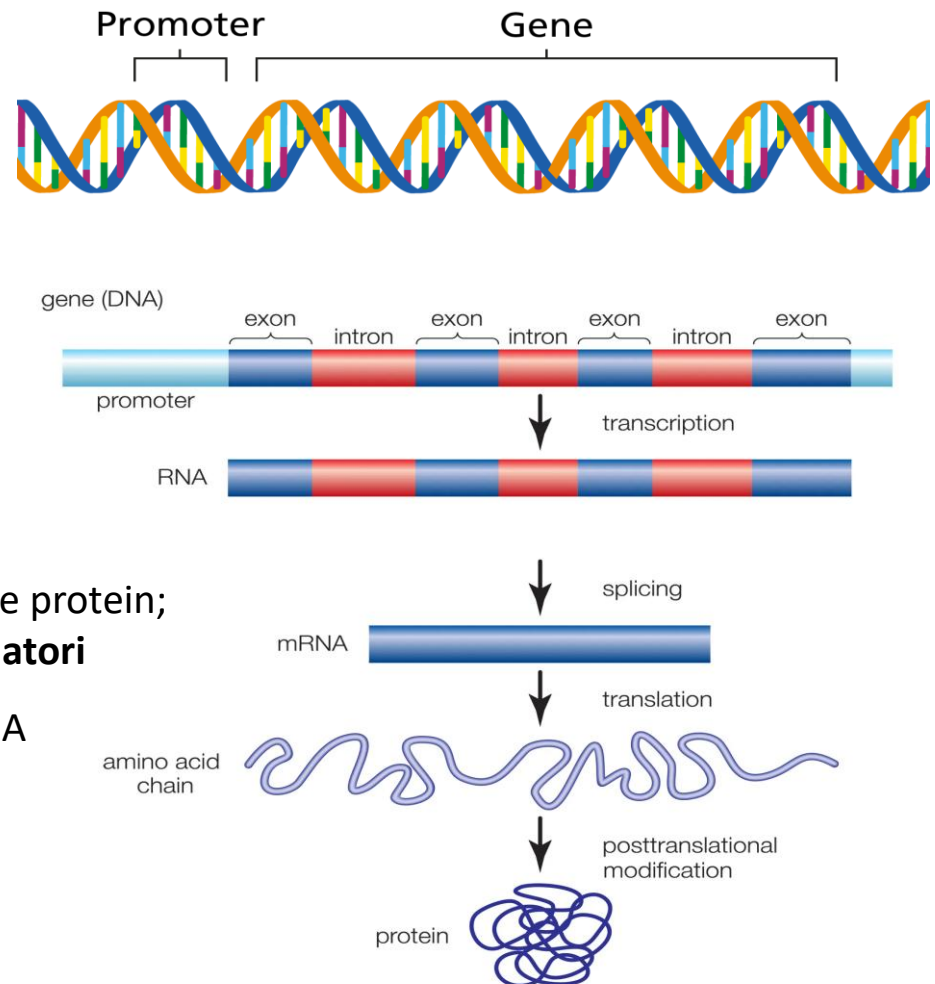
Il PROMOTORE del gene è una regione di DNA costituita da specifiche sequenze nucleotidiche (sequenze consenso), alla quale si lega la proteina RNA polimerasi RNA-Pol per iniziare la trascrizione di un gene, o di più geni.

Sintesi dell'RNA

Maturazione dell'RNA

Gli RNA messaggeri (mRNA) servono per sintetizzare protein;
alter classi di RNA rivestono ruoli **strutturali o regolatori**

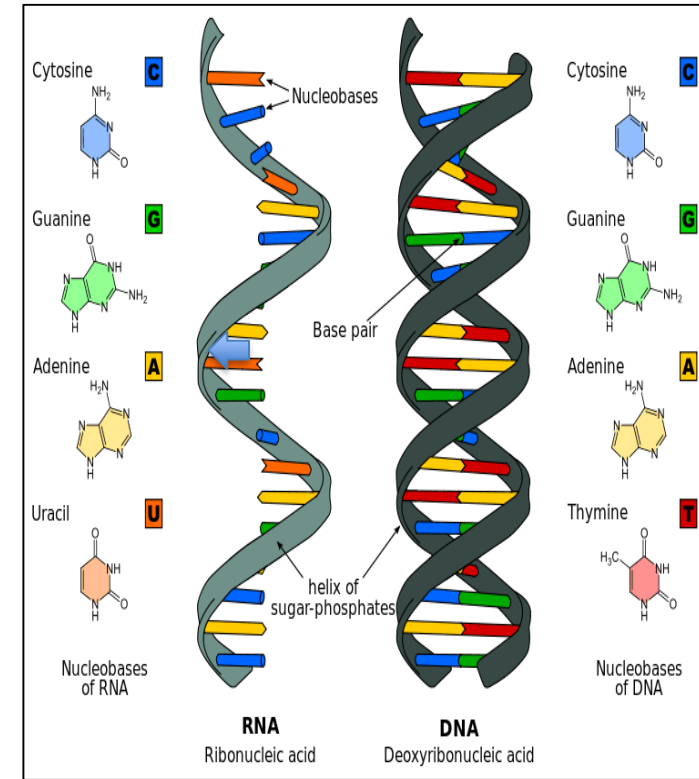
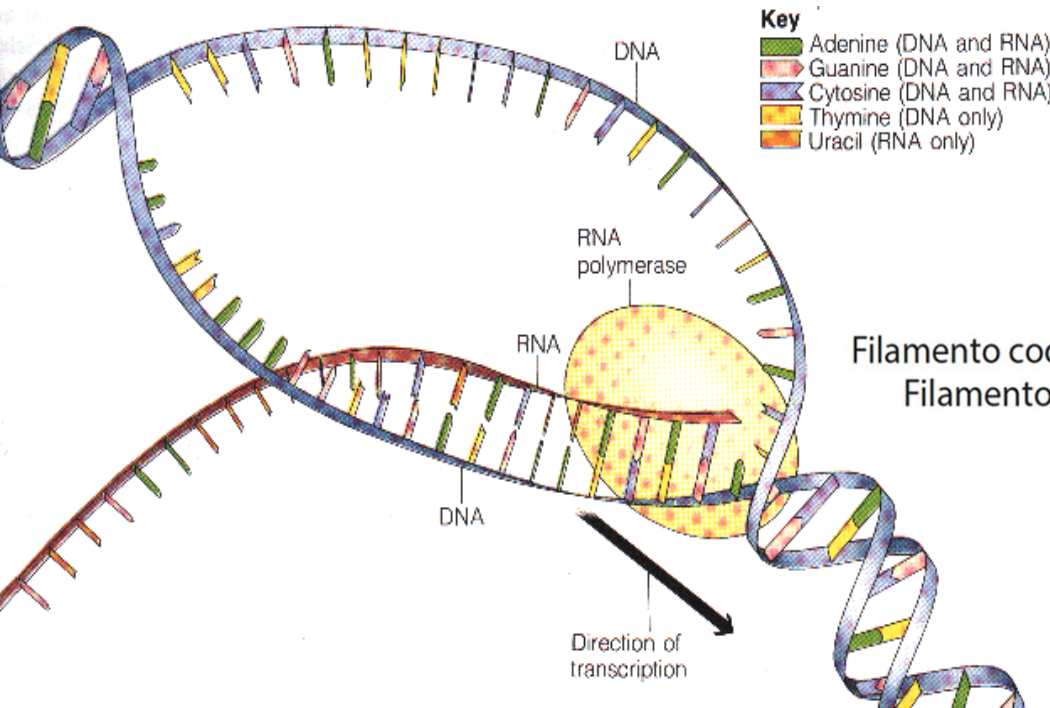
Trasformazione dell'informazione portata dall'mRNA
in una sequenza peptidica



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The Code of Life: La Trascrizione

- **Single-stranded RNA** is produced from a **double-stranded DNA**
- Eukaryotic **RNA polymerase** is a DNA-dependent RNA polymerase as it **uses information from DNA to synthesize a complementary sequence**.
- **Only one strand of the gene is used** as a template for transcription and is referred to as the **template strand**.
- The product is a complementary single-stranded RNA. **RNA polymerase reads DNA 3'-5' (based on the sugar chain bonds)** and produces an RNA molecule complementary to it



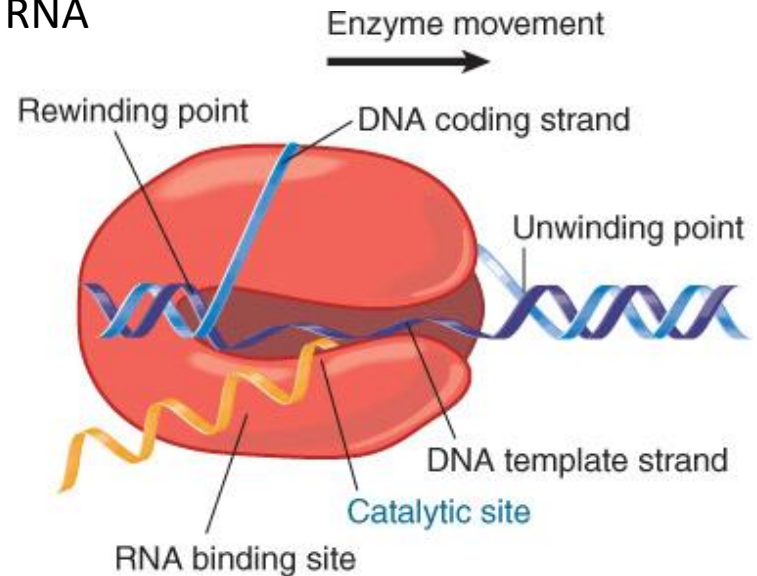
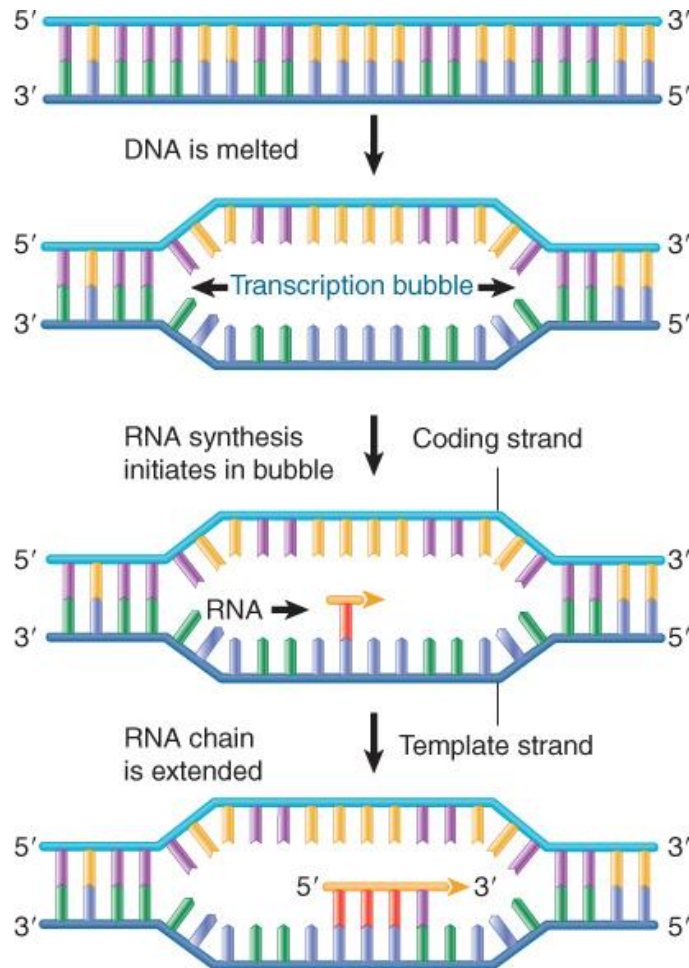
Filamento codificante: 5' — A T G G G C T C C — 3' } DNA
 Filamento stampo: 3' — T A C C C G A G G — 5' }

↓

5' — A U G G G C U C C — 3' mRNA

The Code of Life: La Trascrizione

- **RNA Polymerase II** catalyzes the synthesis of RNA



Each gene is transcribed mostly only on one strand but transcription can proceed in any direction on the chromosome by using different strands for different genes

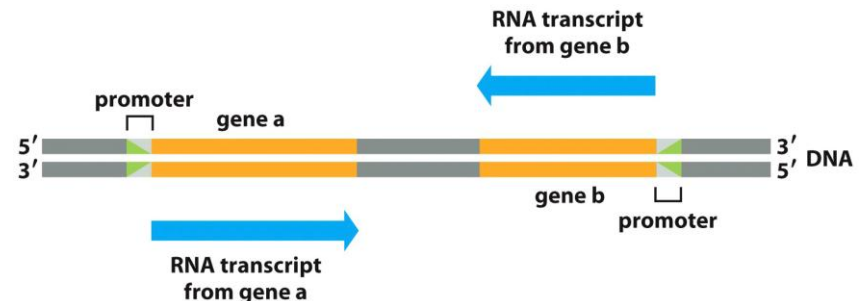
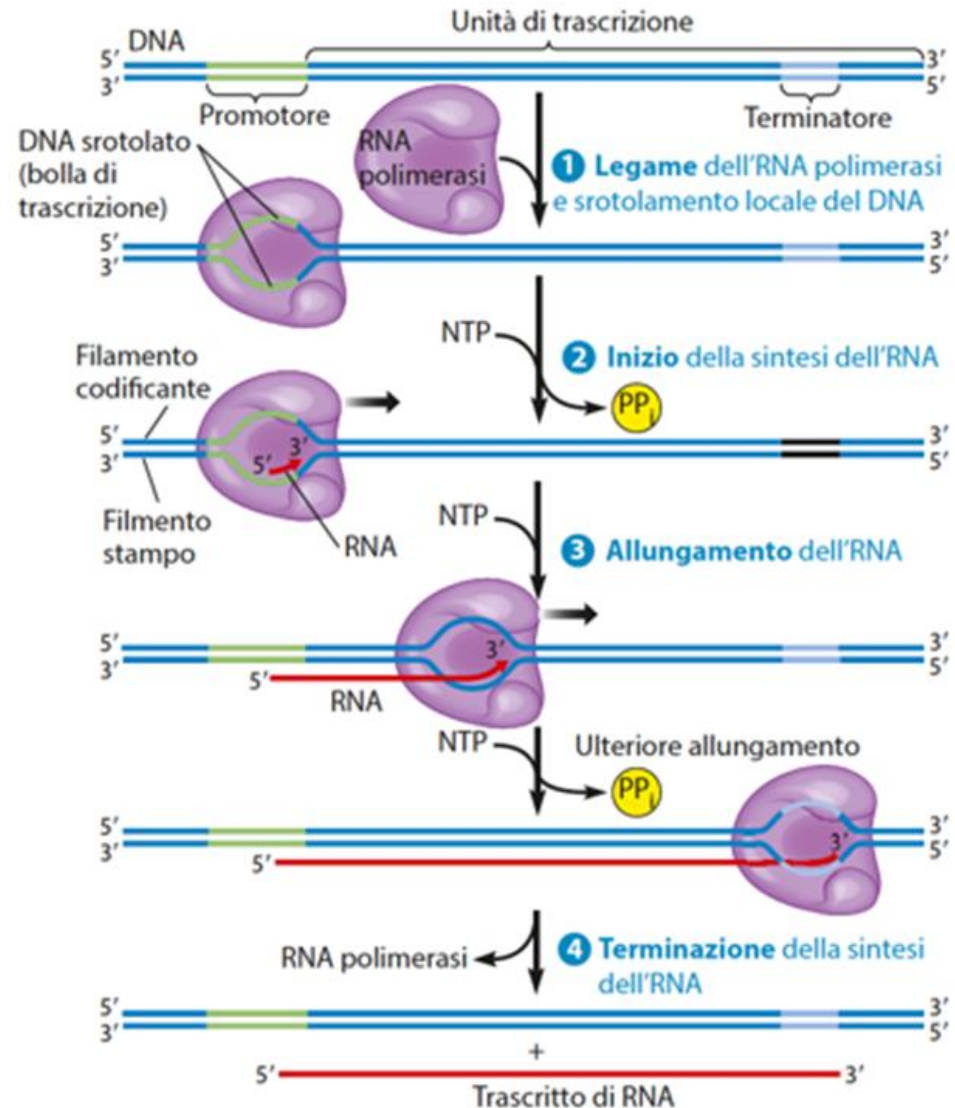


Figure 7-11 Essential Cell Biology, 4th ed. (© Garland Science 2014)

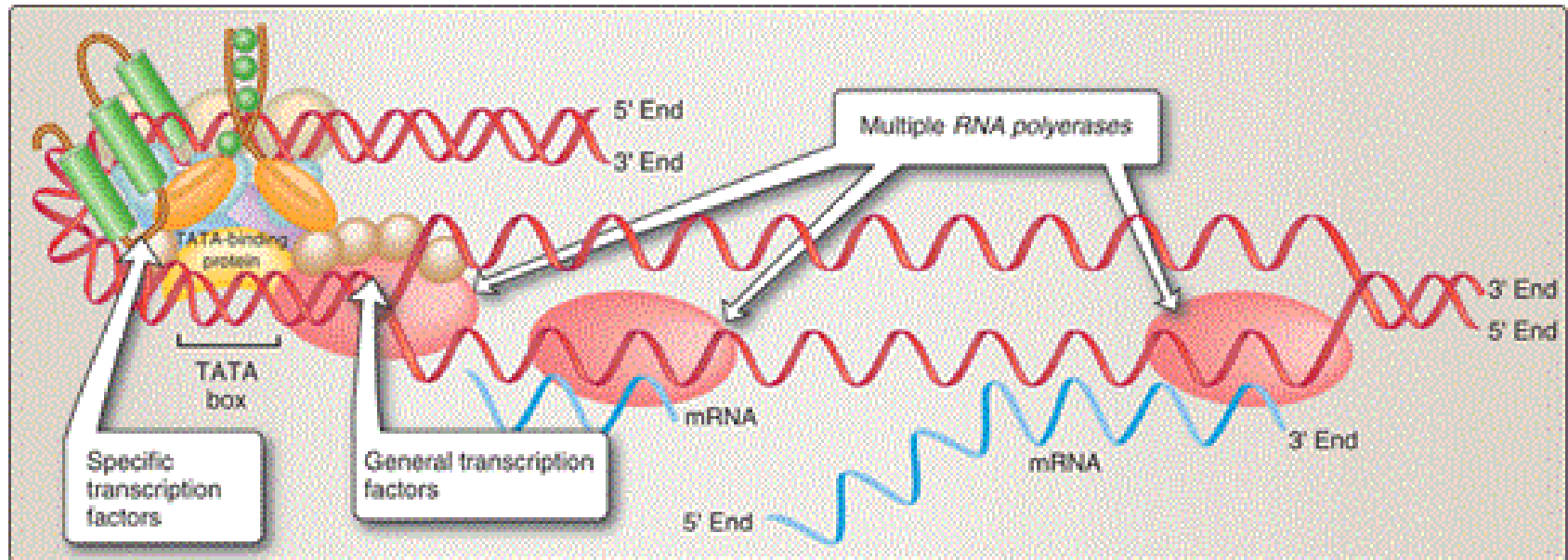
The Code of Life: La Trascrizione

1. Legame della RNA polimerasi ad un promotore
2. Inizio della trascrizione
3. Allungamento
4. Terminazione e rilascio della polimerasi



The Code of Life: La Trascrizione

- **RNA Polymerase II** catalyzes the synthesis of mRNA at a fixed rate of **~ 30-40 nucleotides per second**.
- The **rate of synthesis (transcription)** can be **modulated**, and the **number of polymerases** that simultaneously synthesize RNA from a given gene determine the absolute transcription rate.
- Specific **molecular events modulate the number of RNA polymerase** molecules actively synthesizing RNA



The Code of Life: Categories of RNA Molecules

TABLE 6-1 Principal Types of RNAs Produced in Cells

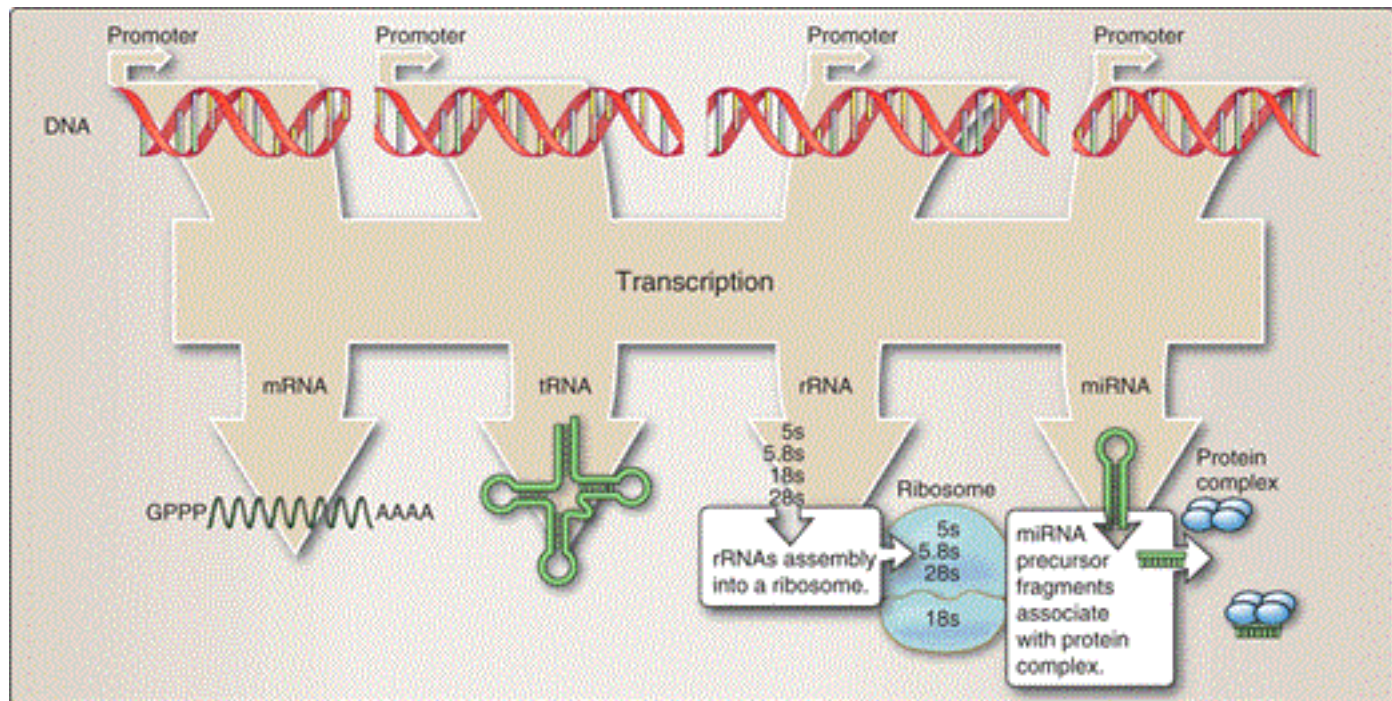
Type of RNA	Function
mRNAs	Messenger RNAs, code for proteins
rRNAs	Ribosomal RNAs, form the basic structure of the ribosome and catalyze protein synthesis
tRNAs	Transfer RNAs, central to protein synthesis as adaptors between mRNA and amino acids
snRNAs	Small nuclear RNAs, function in a variety of nuclear processes, including the splicing of pre-mRNA
snoRNAs	Small nucleolar RNAs, help to process and chemically modify rRNAs
miRNAs	MicroRNAs, regulate gene expression by blocking translation of specific mRNAs and cause their degradation
siRNAs	Small interfering RNAs, turn off gene expression by directing the degradation of selective mRNAs and the establishment of compact chromatin structures
piRNAs	Piwi-interacting RNAs, bind to piwi proteins and protect the germ line from transposable elements
lncRNAs	Long noncoding RNAs, many of which serve as scaffolds; they regulate diverse cell processes, including X-chromosome inactivation

Table 6-1 Molecular Biology of the Cell 6e (© Garland Science 2015)

The Code of Life: RNA polymerases and RNA classes

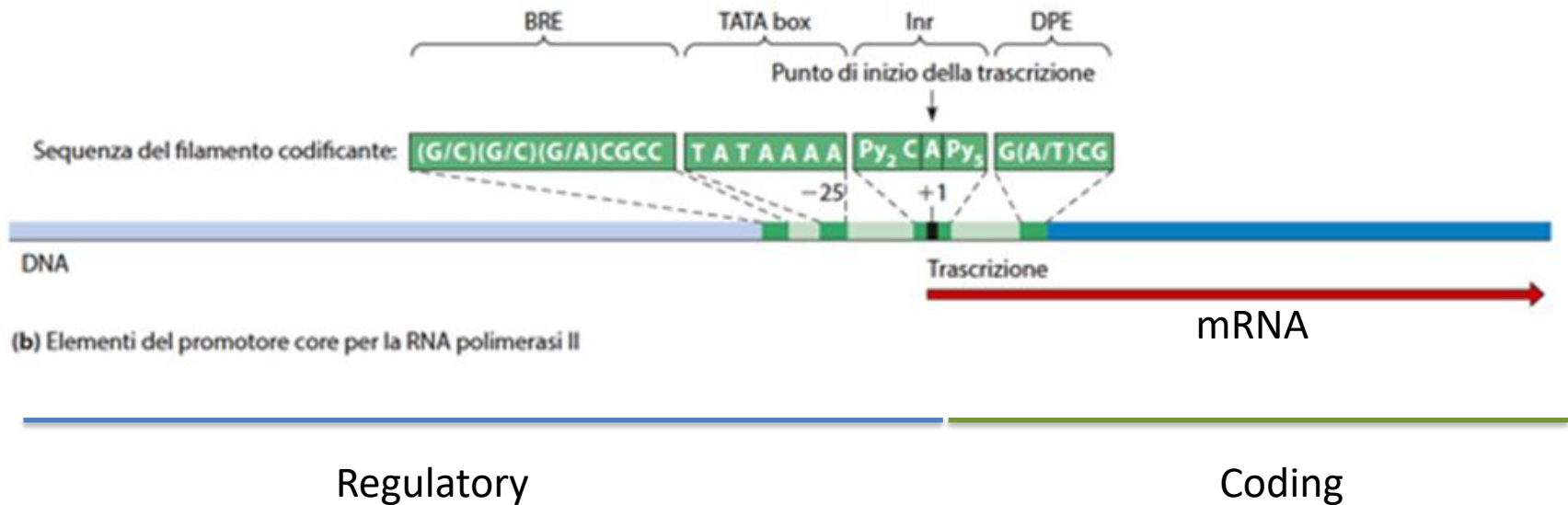
There are several distinct RNA polymerases in eukaryotic cells:

- **RNA polymerase I** synthesizes most **rRNAs** (5.8S, 18S, 28S) involved in facilitating protein synthesis by the **ribosome**.
- **RNA polymerase II** is responsible for the synthesis of **mRNA**, but also for **long-non coding RNAs** and **miRNAs**.
- **RNA polymerase III** catalyzes the synthesis of **tRNAs** and **5S rRNAs** and other small RNAs.



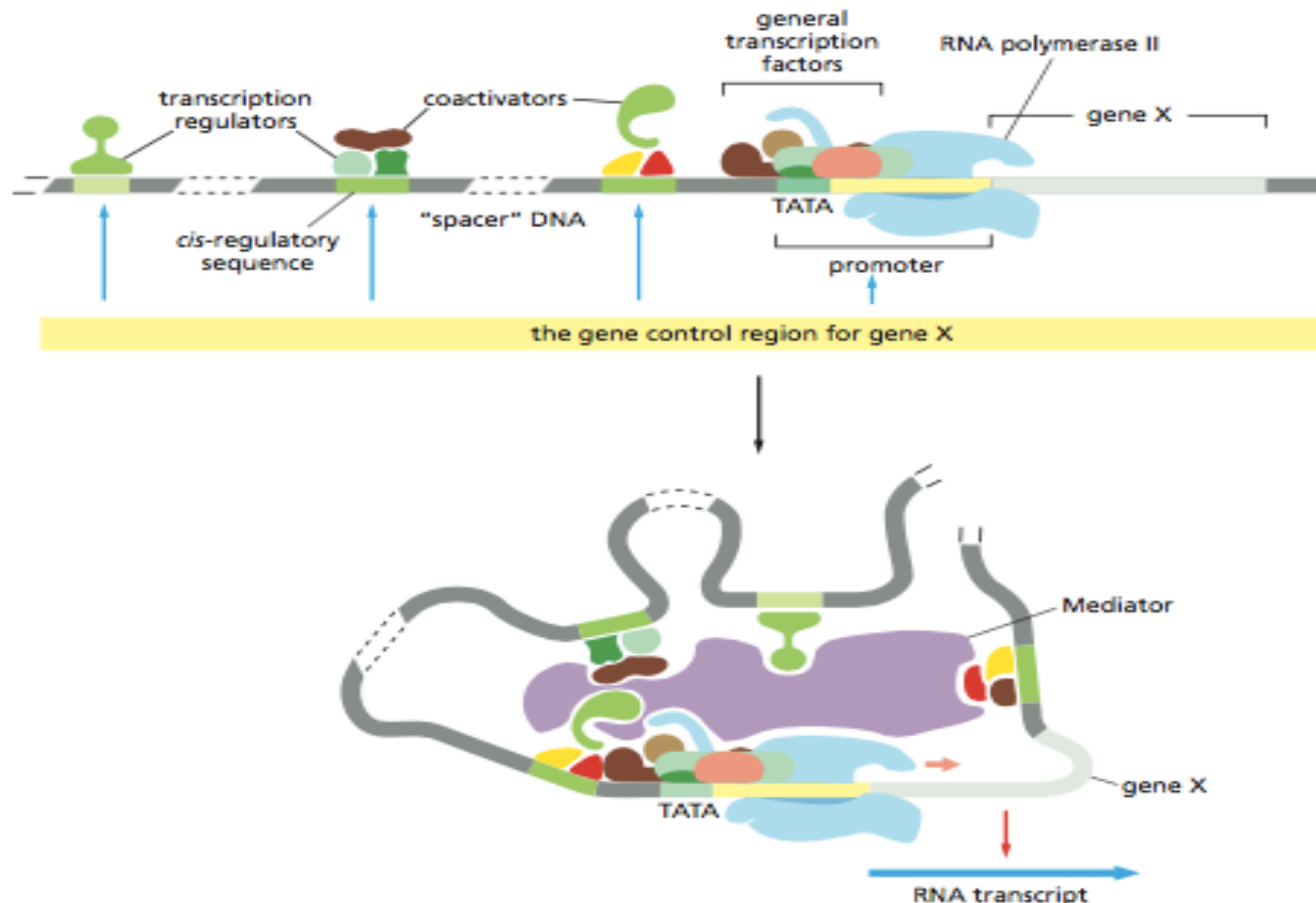
The Code of Life: Regulation of transcription

An mRNA-producing **eukaryotic gene** can be divided into its regulatory and coding regions as defined by **the transcription start site**.



The Code of Life: Regulatory regions

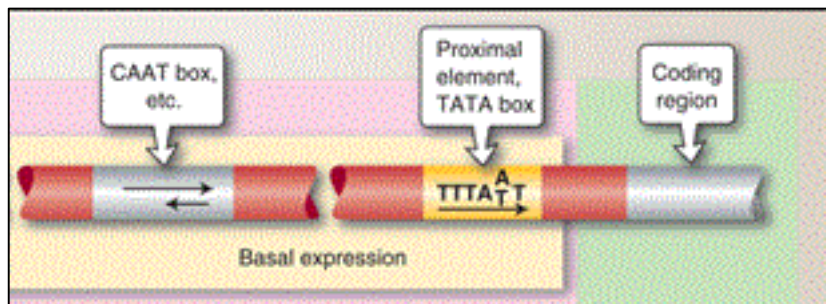
The **regulatory region** of a gene consists of **two classes of sequences** respectively ensuring **basal expression** and **regulated expression**.



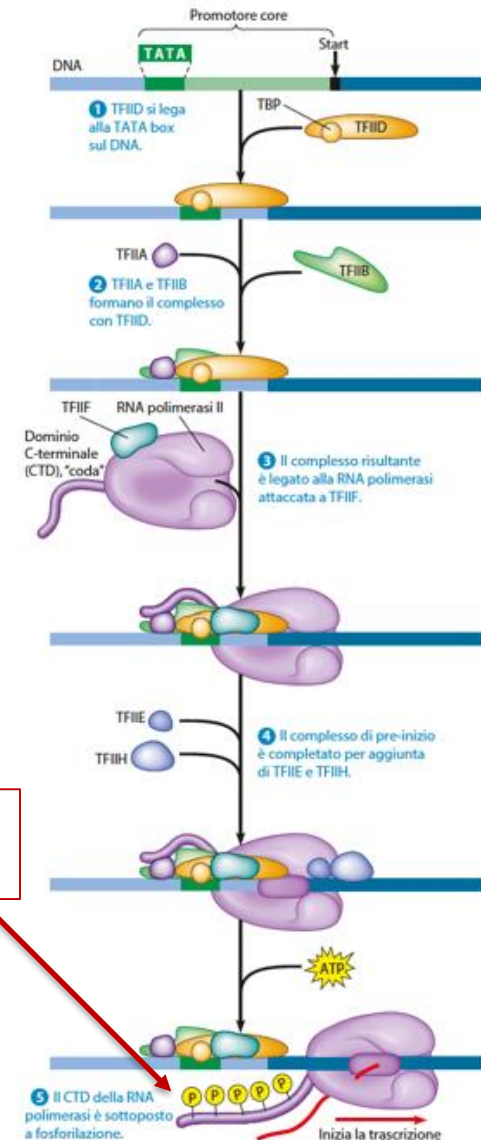
The Code of Life: Generic Transcription Factors and Basal Expression

Basal expression

- **Basal responsive elements (BRE)** generally have two components:
 - **proximal component** (generally the **TATA box**), directs RNA polymerase II to the correct site to **ensure fidelity of initiation**
 - **distal component** specifies the **frequency of initiation** (for example CAAT or GC boxes).
- **These boxes bind specific proteins: general transcription factors**



RNA-POLII Activation



The Code of Life: RNA Polymerase II Requires a Set of General Transcription Factors

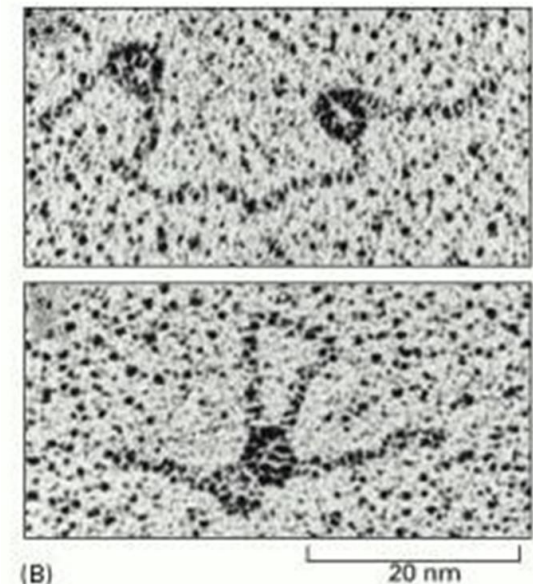
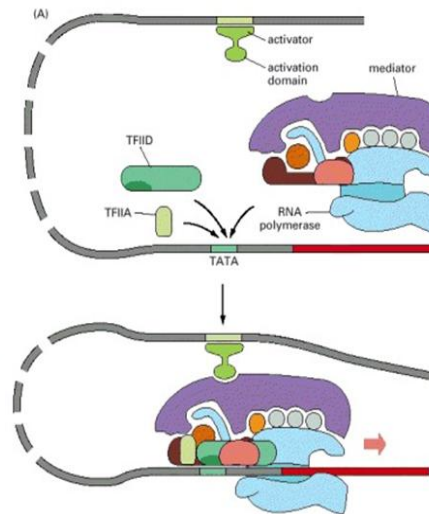
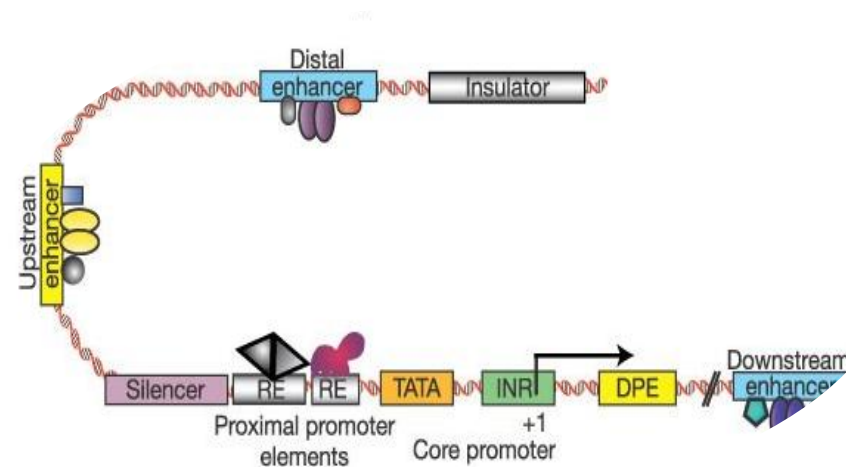
TABLE 6–3 The General Transcription Factors Needed for Transcription Initiation by Eukaryotic RNA Polymerase II

Name	Number of subunits	Roles in transition initiation
TFIID TBP subunit TAF subunits	1 ~11	Recognizes TATA box Recognizes other DNA sequences near the transcription start point; regulates DNA-binding by TBP
TFIIB	1	Recognizes BRE element in promoters; accurately positions RNA polymerase at the start site of transcription
TFIIF	3	Stabilizes RNA polymerase interaction with TBP and TFIIB; helps attract TFIIIE and TFIIH
TFIIIE	2	Attracts and regulates TFIIH
TFIIH	9	Unwinds DNA at the transcription start point, phosphorylates Ser5 of the RNA polymerase CTD; releases RNA polymerase from the promoter
TFIID is composed of TBP and ~11 additional subunits called TAFs (TBP-associated factors); CTD, C-terminal domain.		

Table 6-3 Molecular Biology of the Cell 6e (© Garland Science 2015)

The Code of Life: Regulation of Transcription

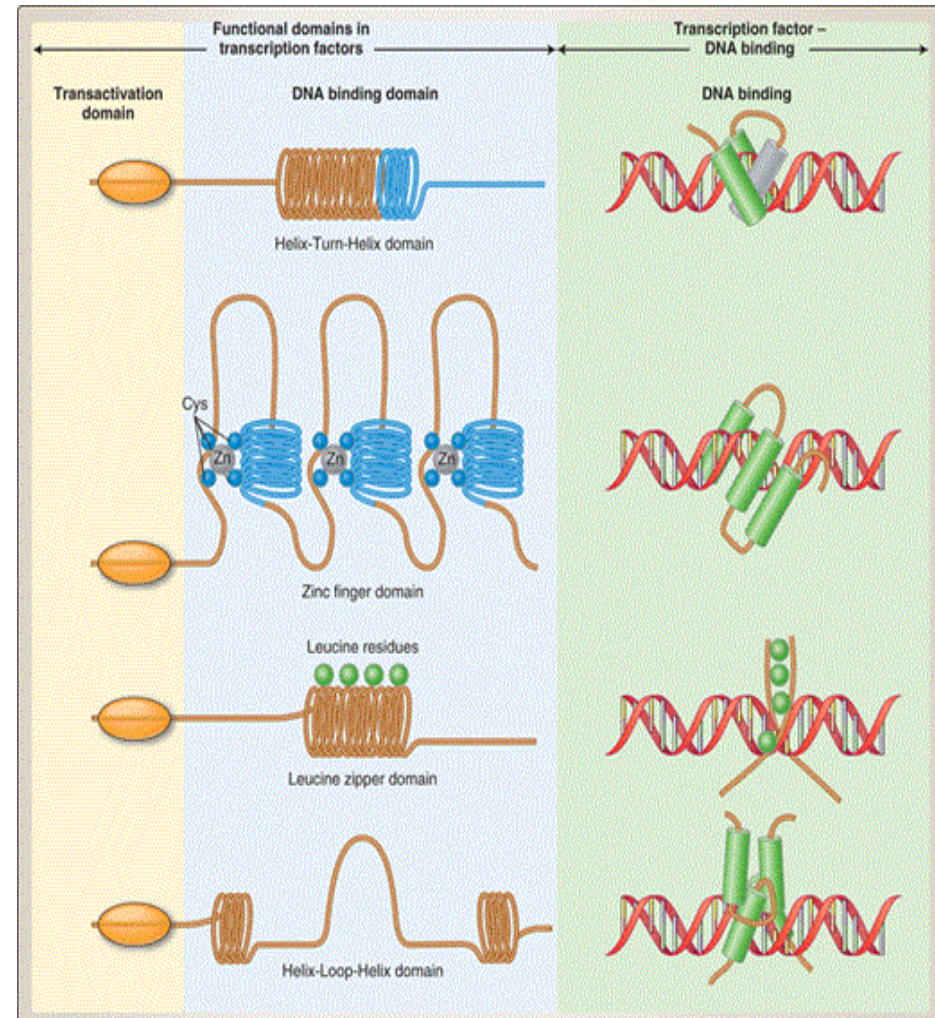
- **Enhancers and response elements** are also **recognized by specific Transcription Factors** that regulate gene expression in response to molecular signals (DNA damage, hormones, etc.).
- They increase or decrease the initiation rate of transcription.
- **Both upstream and downstream** from the transcription start site. They can exert their effects even when **located hundreds or even thousands of bases away** from the transcription start site.
- They may function in an **orientation-independent fashion**.



The Code of Life: Transcription Factors

- **Transcription factors** are proteins that have a modular design consisting of at least **two distinct domains**:
 - **DNA-binding** domain consists of the structural motif that recognizes specific DNA sequences (DNA binding)
 - **Transcription-activating domain** contacts the transcriptional machinery and accelerates the rate of transcription initiation by accelerating the assembly of the general transcription factors at the promoter site (transcription activating)

Different types of DNA-binding domains



The Code of Life: Specific Transcription Factors

DNA-binding domain of the transcription factors consists of a structural motif that recognizes specific stretches of nucleotides (DNA binding)

Interazioni AA-Nucleotide
sono legami non-covalenti

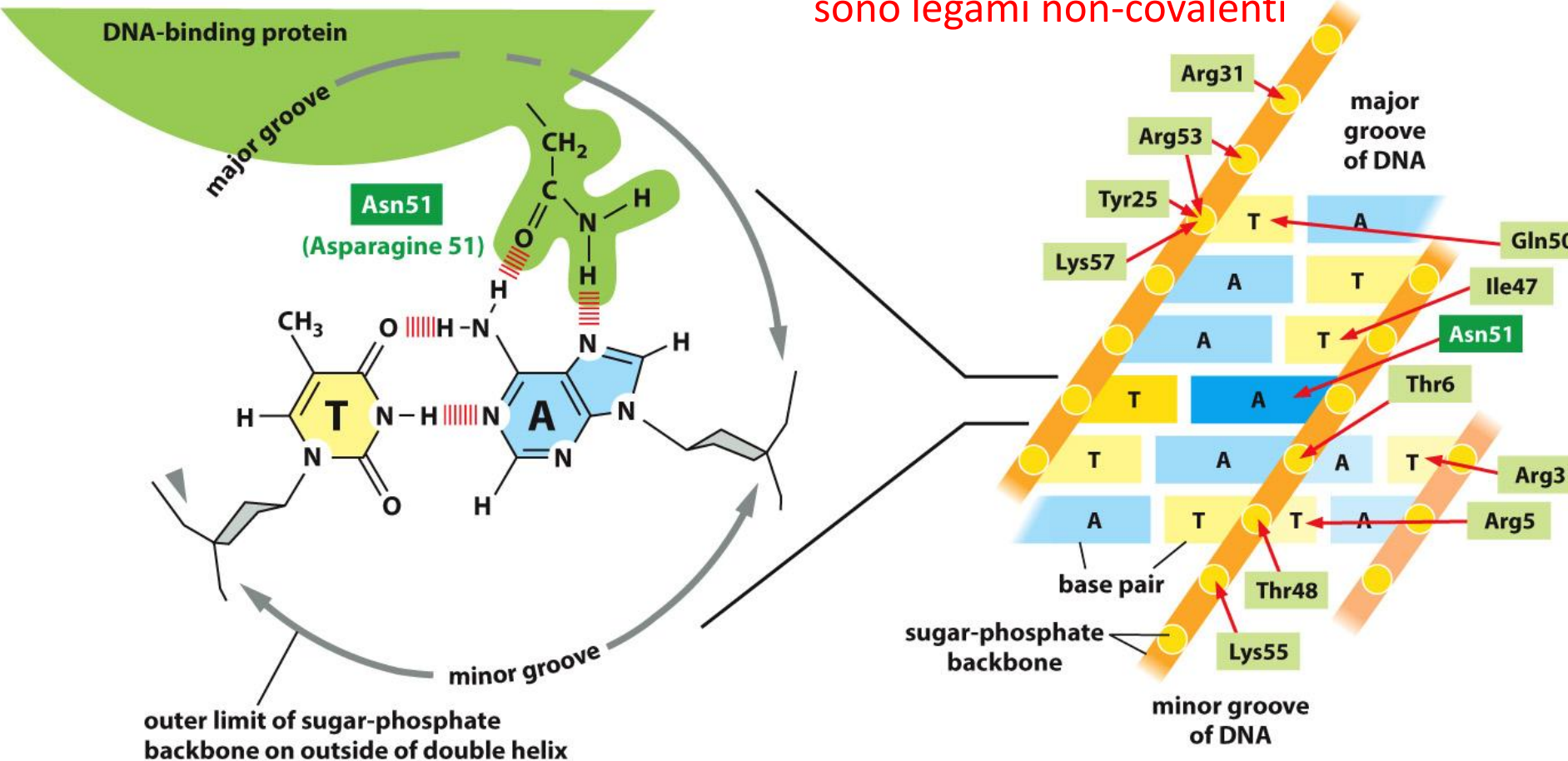


Figure 7-8 Molecular Biology of the Cell 6e (© Garland Science 2015)

Consensus sequences recognized by transcription factors

- Transcription factors bind to the consensus sequences “in cis” (same chromosome = in cis) to the genes they control

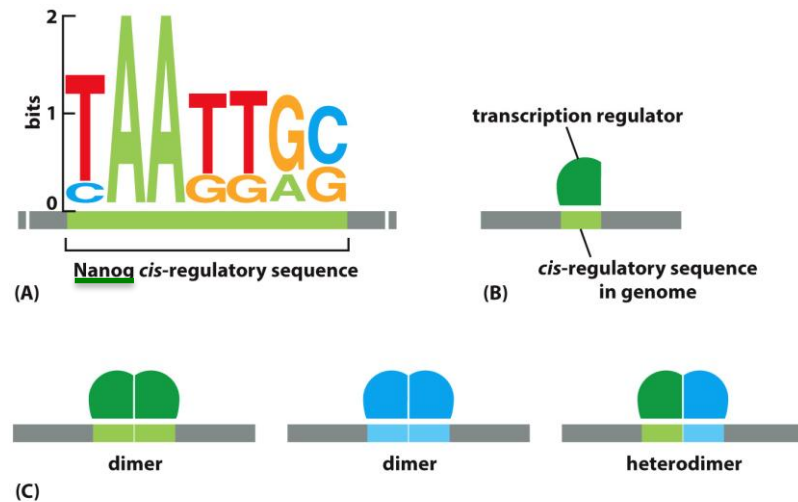


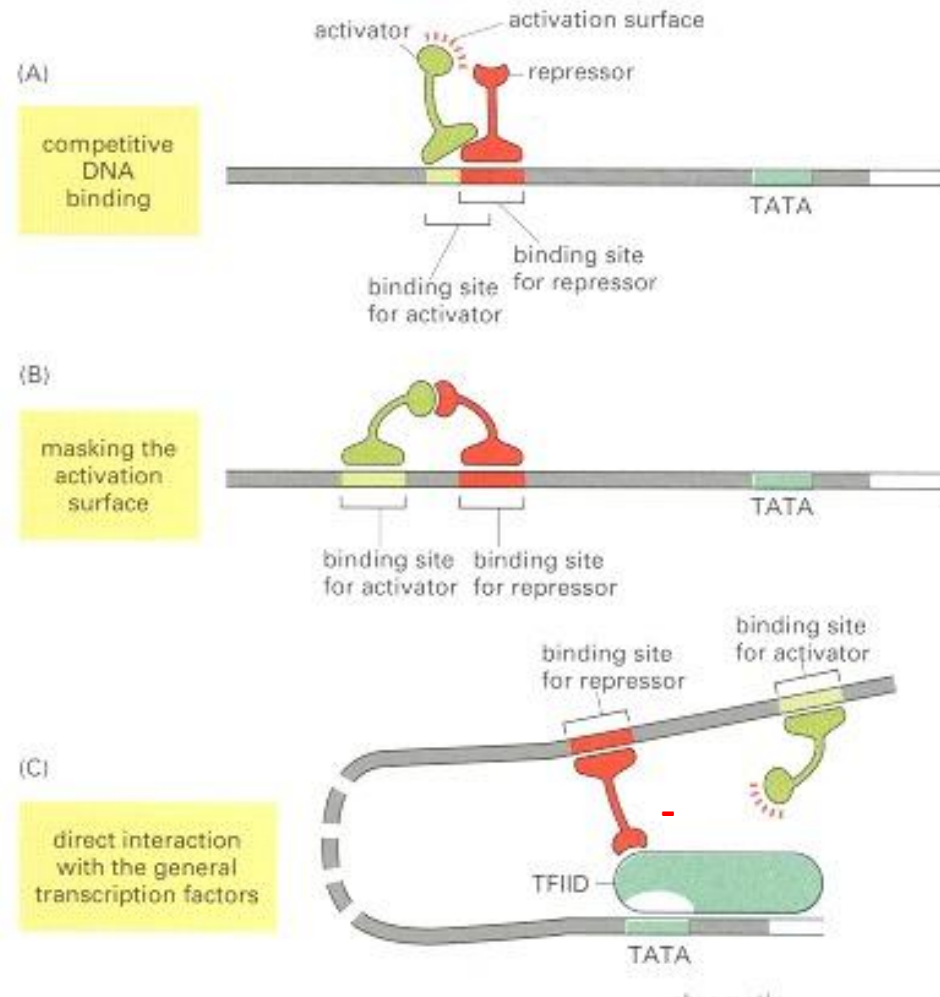
Figure 7-9 Molecular Biology of the Cell 6e (© Garland Science 2015)

- Different transcription factors have different sequence binding specificity**

Transcription Factor	Sequence Recognized
Myc and Max	CACGTG
Fos and Jun	TGACTCA
TR (thyroid hormone receptor)	GTGTCAAAGGTCA
MyoD	CAACTGAC
RAR (retinoic acid receptor)	ACGTCATGACCT

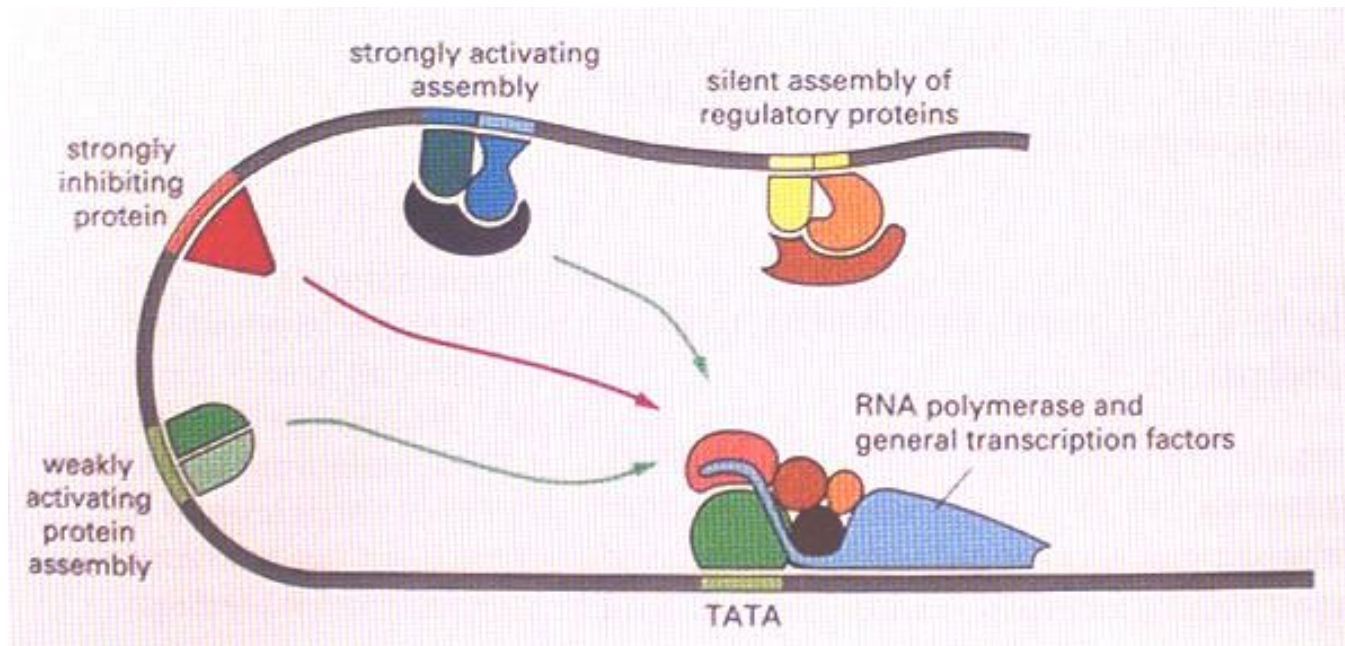
The Code of Life: Transcriptional repressors

Repressor **transcription factors** can operate by:



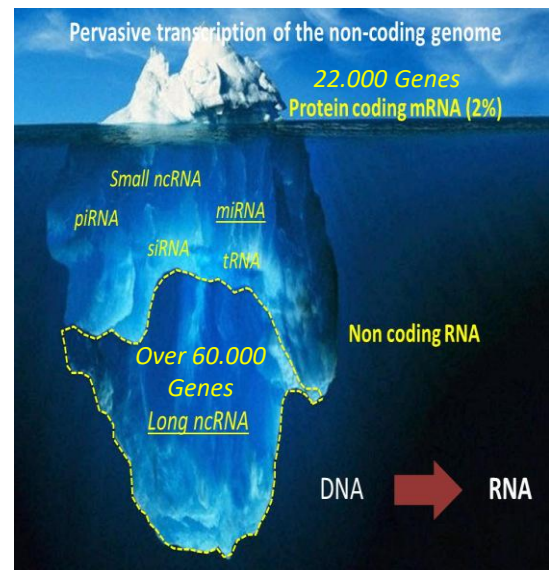
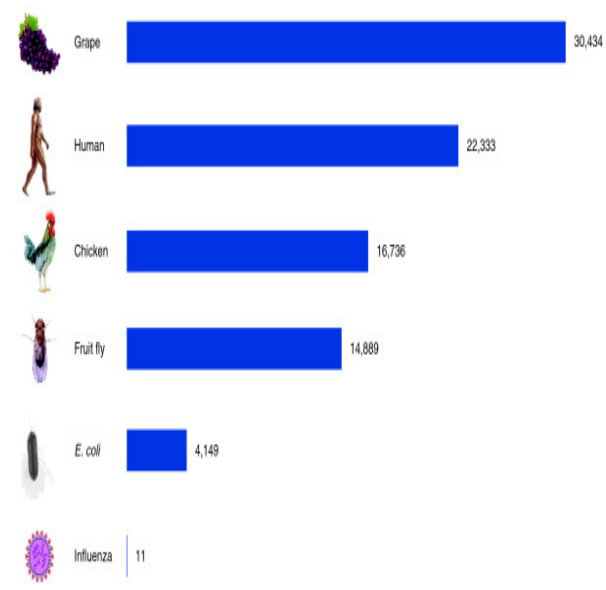
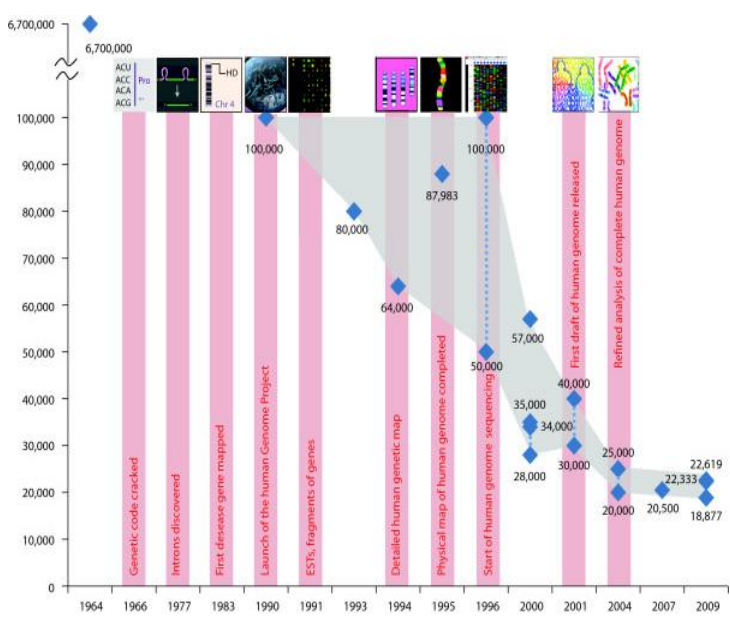
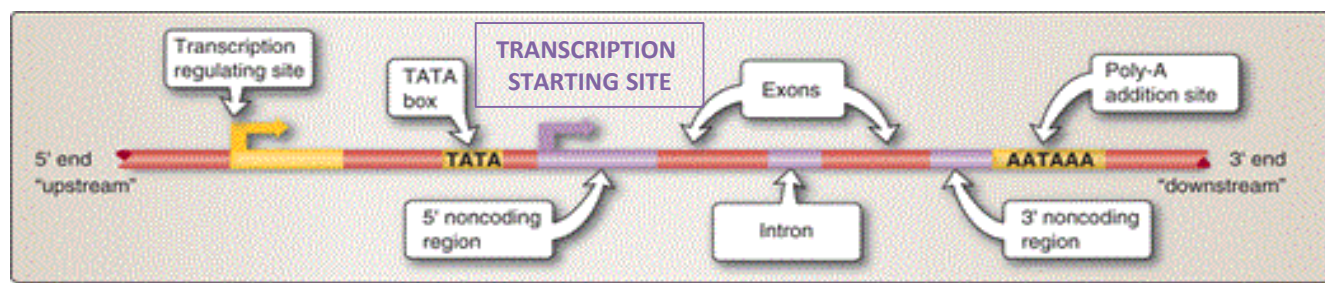
The Code of Life: Transcriptional Activation vs Repression

Eucaryotic gene expression/repressor depends on the combination of stimulatory and repressive actions operate by different Transcription Factors.



The Code of Life: Gene Structure in Eukariotes

- Genes are formed by **untranscribed (controlling) & transcribed** regions.
- Eukaryotic genes are composed of **exons, introns**, and **5' and 3' untranslated regions (UTRs)**.



The Code of Life: Genes Organization

Length and number of exons/introns varies among genes

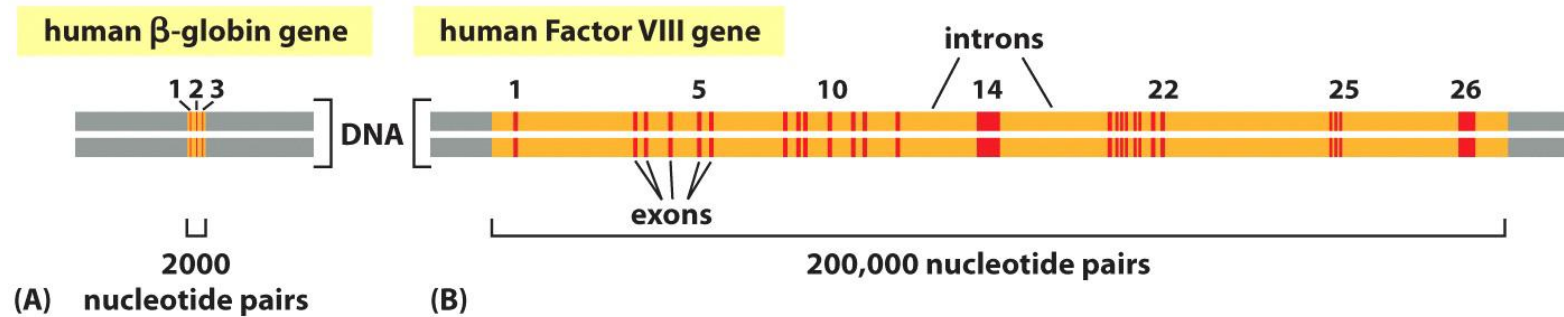


Figure 7-18 Essential Cell Biology, 4th ed. (© Garland Science 2014)

TABLE 4-1 Some Vital Statistics for the Human Genome	
Human genome	
DNA length	3.2×10^9 nucleotide pairs*
Number of genes coding for proteins	Approximately 21,000
Largest gene coding for protein	2.4×10^6 nucleotide pairs
Mean size for protein-coding genes	27,000 nucleotide pairs
Smallest number of exons per gene	1
Largest number of exons per gene	178
Mean number of exons per gene	10.4
Largest exon size	17,106 nucleotide pairs
Mean exon size	145 nucleotide pairs

The Code of Life: Intron Length in Evolution

Long introns are more frequent in human genes than other eukariotes

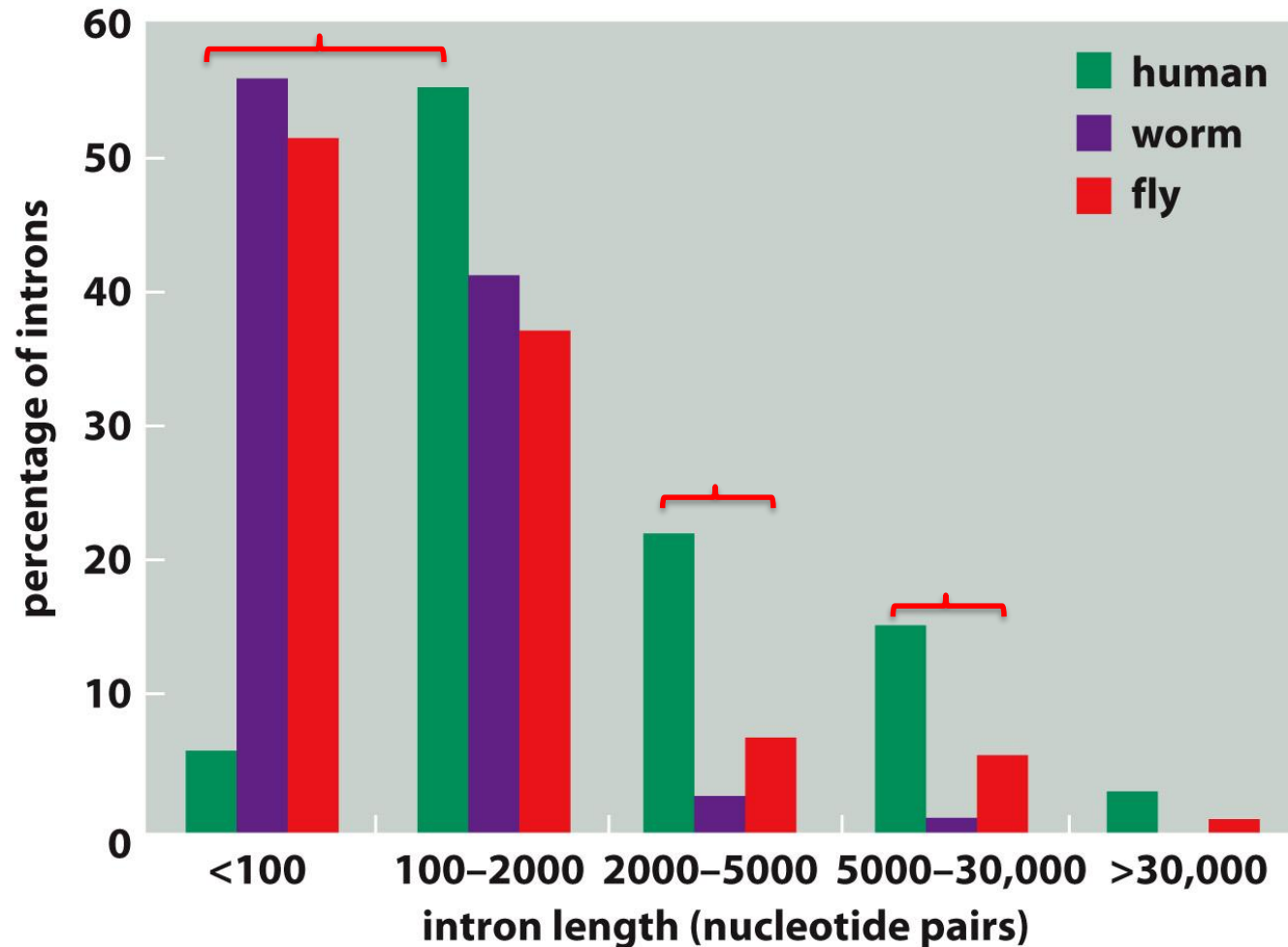
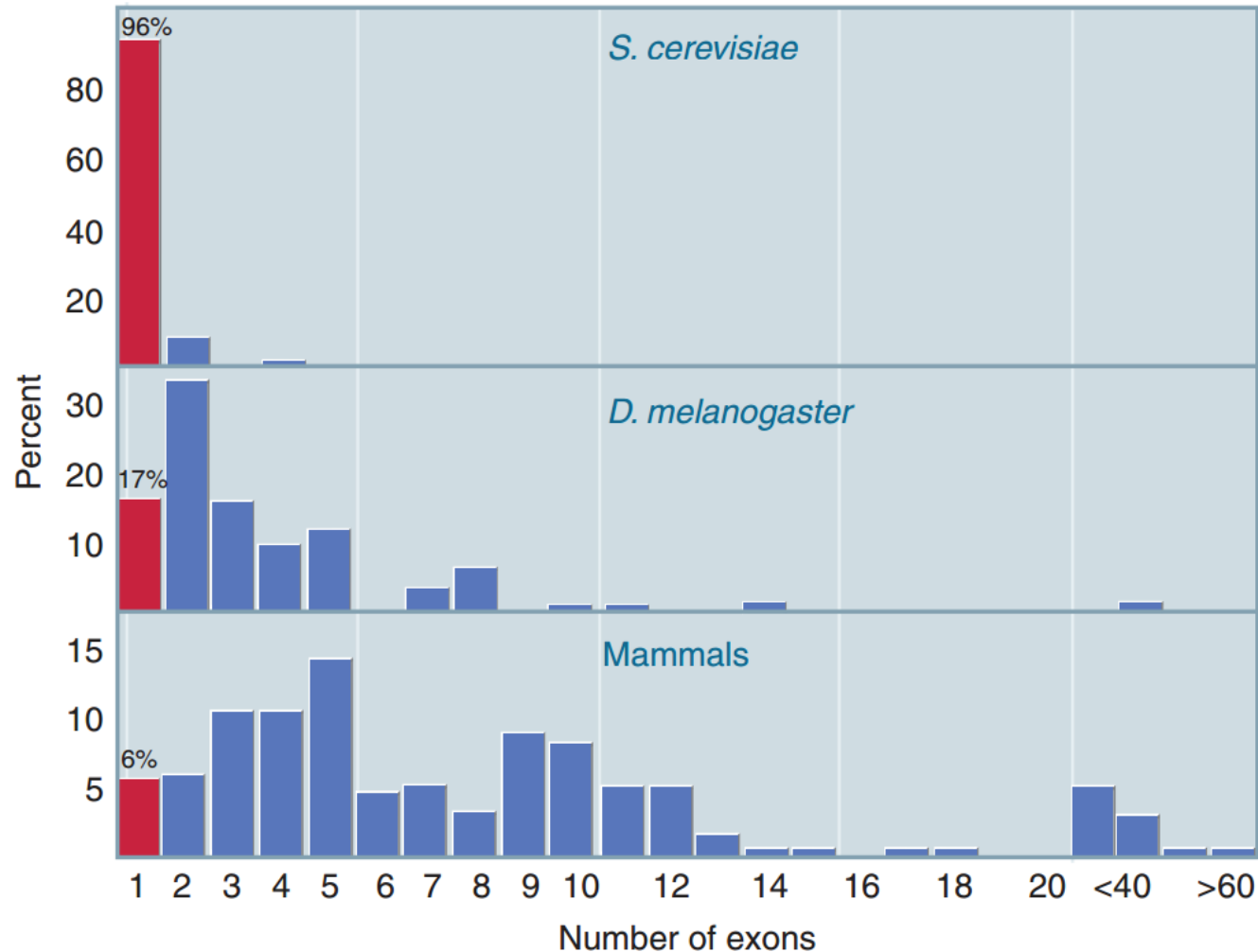


Figure 6-31b Molecular Biology of the Cell 6e (© Garland Science 2015)

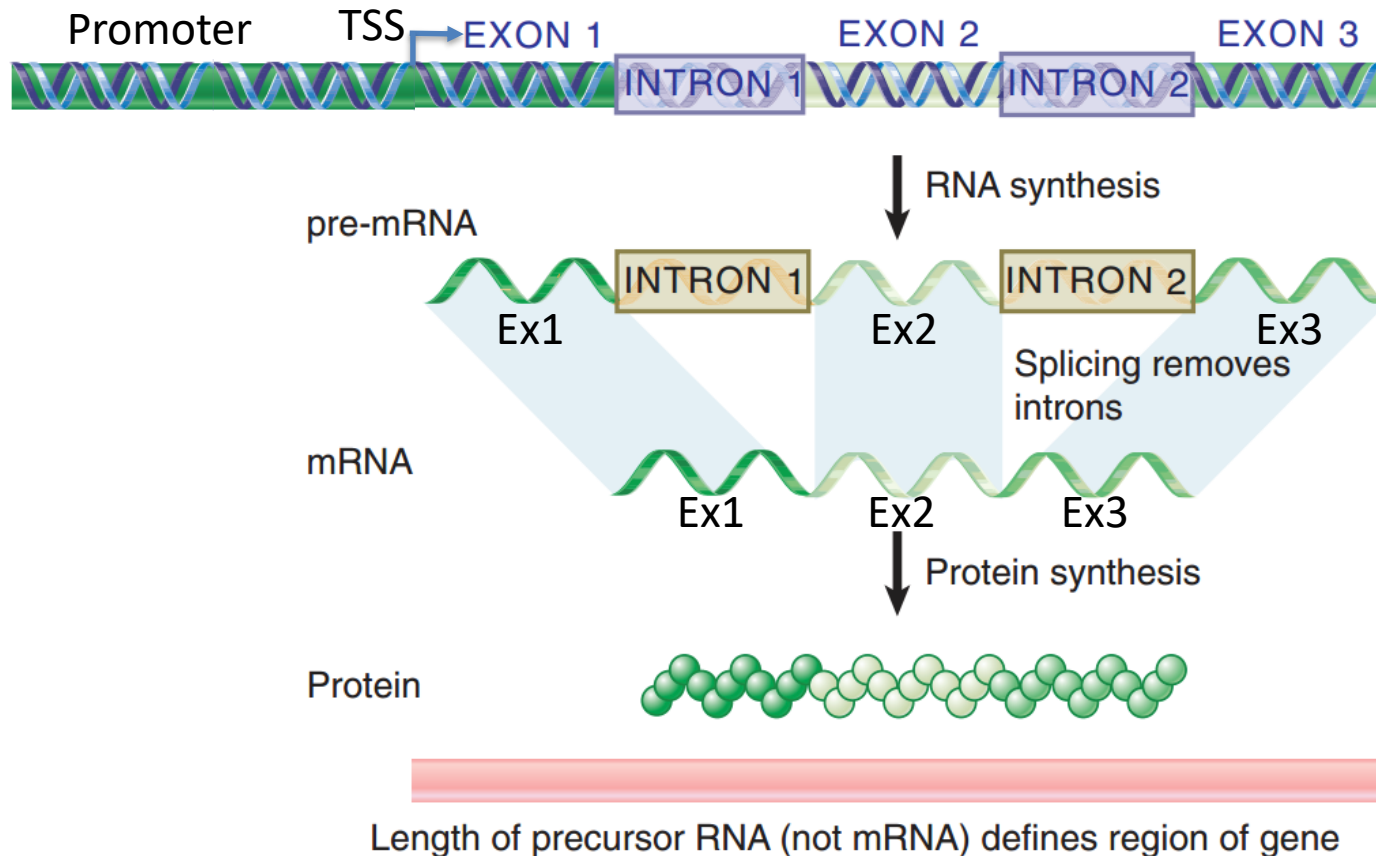
The Code of Life: Intron Length in Evolution

Number of Exons (and introns) increases with the evolution



The Code of Life: Exons/Introns in Eukariotes

- **Exons** are the sequences **retained in the mature mRNA** product
- **Introns** are the intervening sequences that are **removed** when the primary mRNA transcript is processed into the mature mRNA
- The process of intron removal is called **SPLICING**



The Code of Life: mRNA Maturation-Splicing

- The **intron segments are removed** and the exons are joined at specific sequences, called **donor and acceptor sites**, to form the mature mRNA by a mechanism called **splicing**
- The splicing machinery is composed by ribonucleoproteins combined with the primary transcript to form the **spliceosome**
- Spliceosomes** comprise:
 - **primary transcript**
 - **5 small nuclear RNAs (U1, U2, U5, and U4/6)**
 - **> 200 proteins, collectively called snRNPs**
(small nuclear ribonuclear proteins, pronounced "snurps")

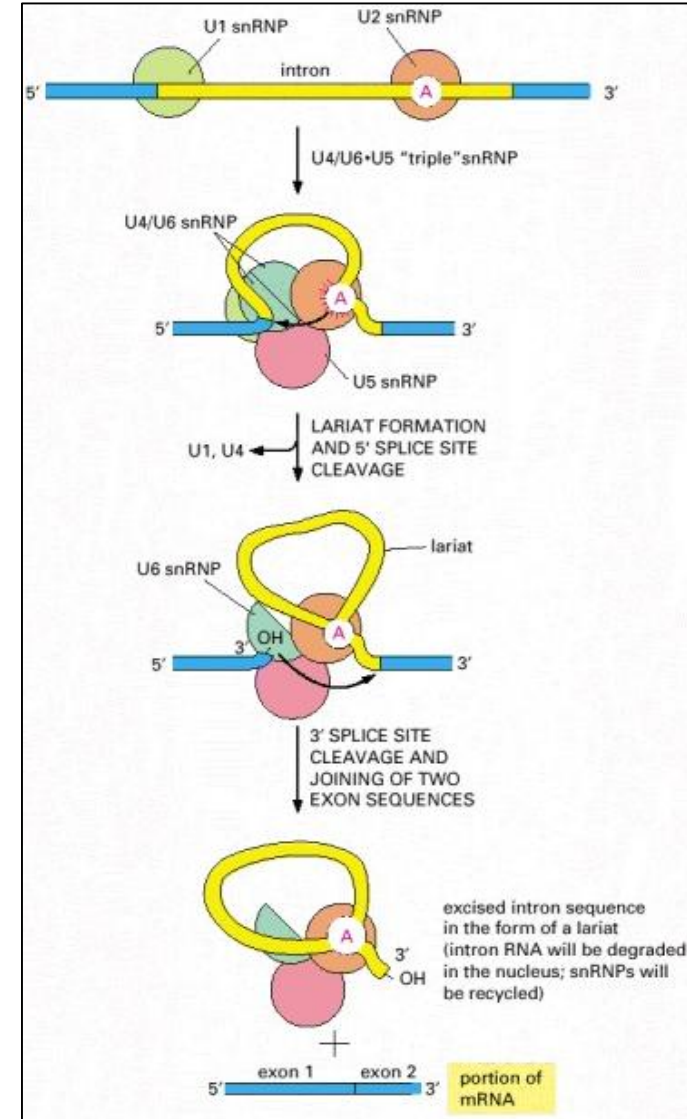
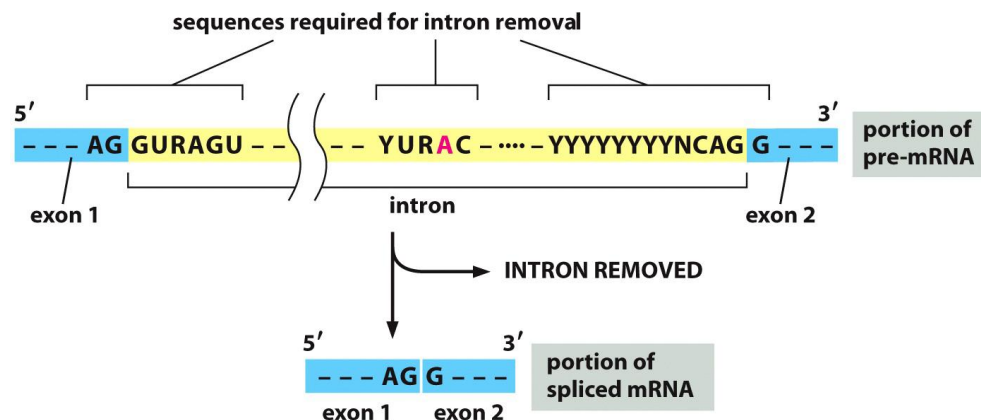
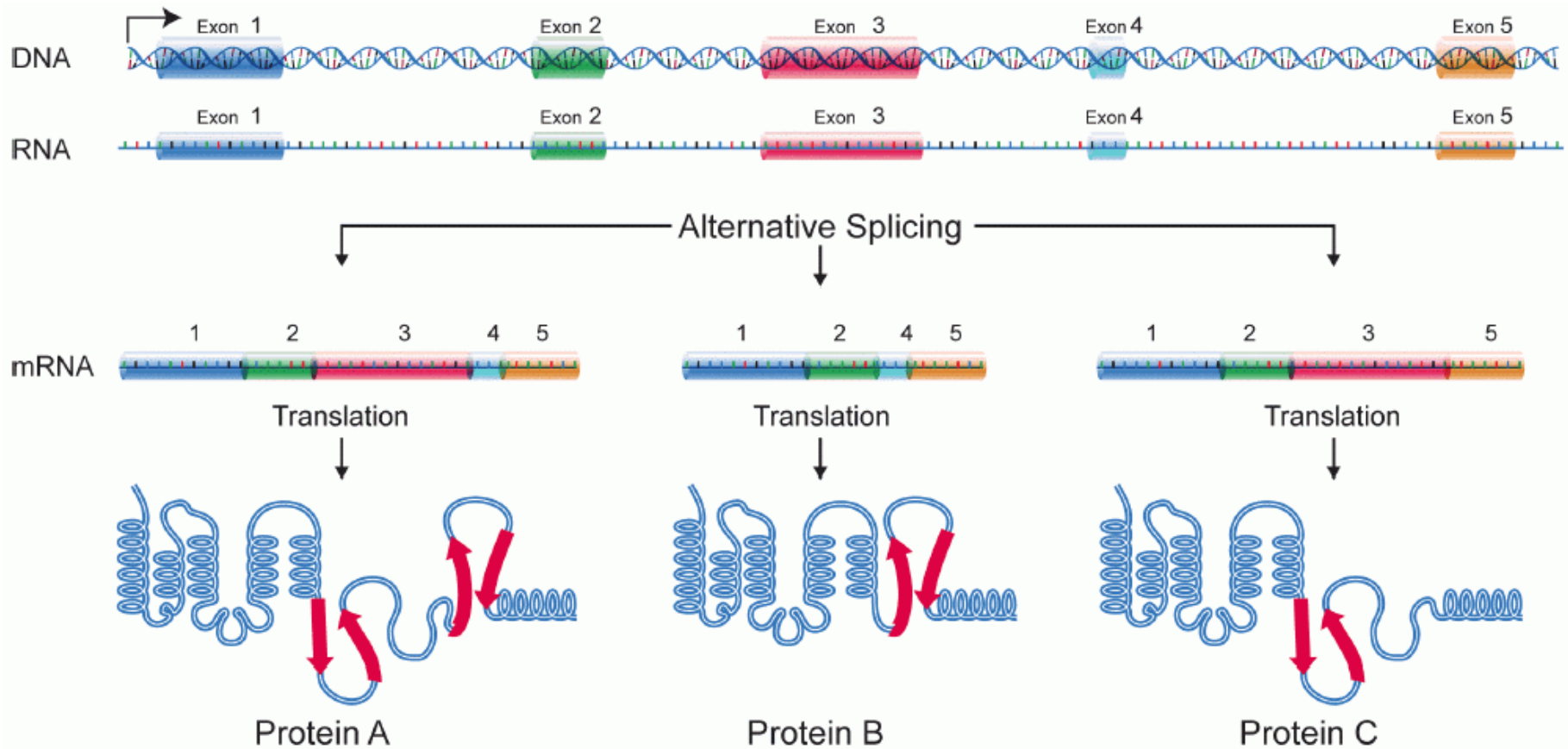


Figure 7-19 Essential Cell Biology, 4th ed. (© Garland Science 2014)

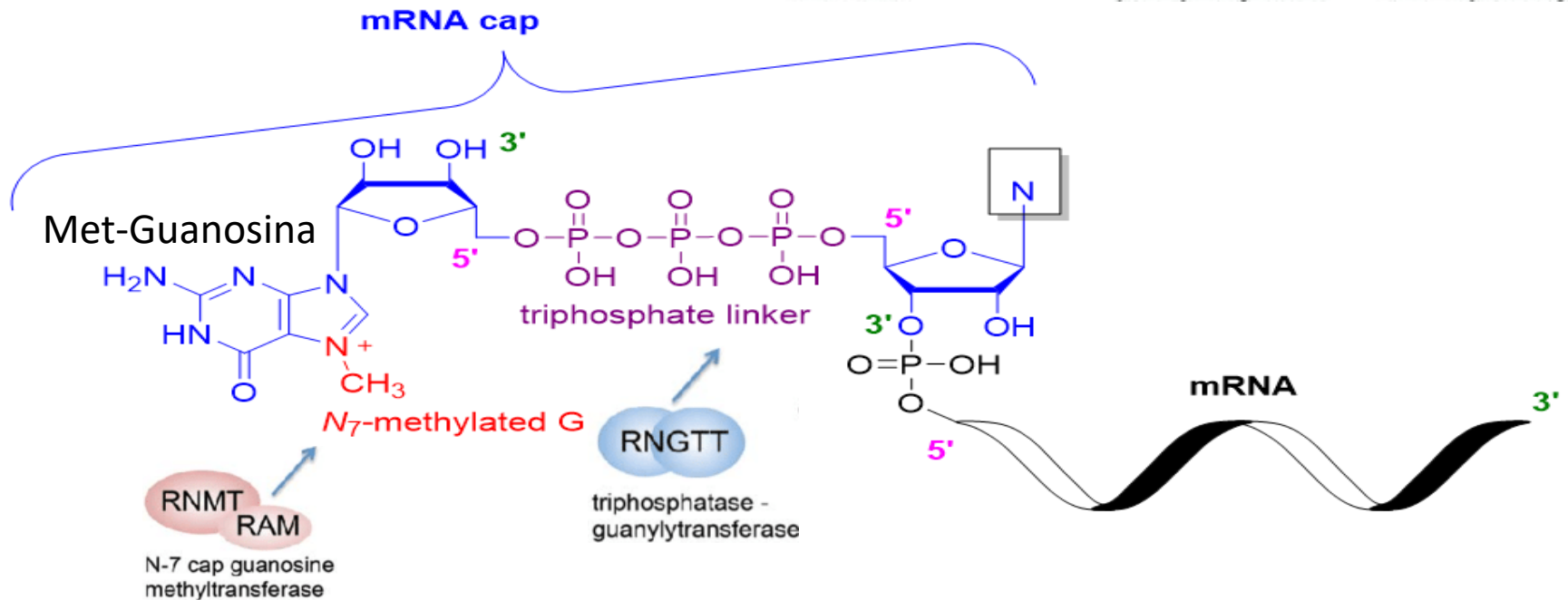
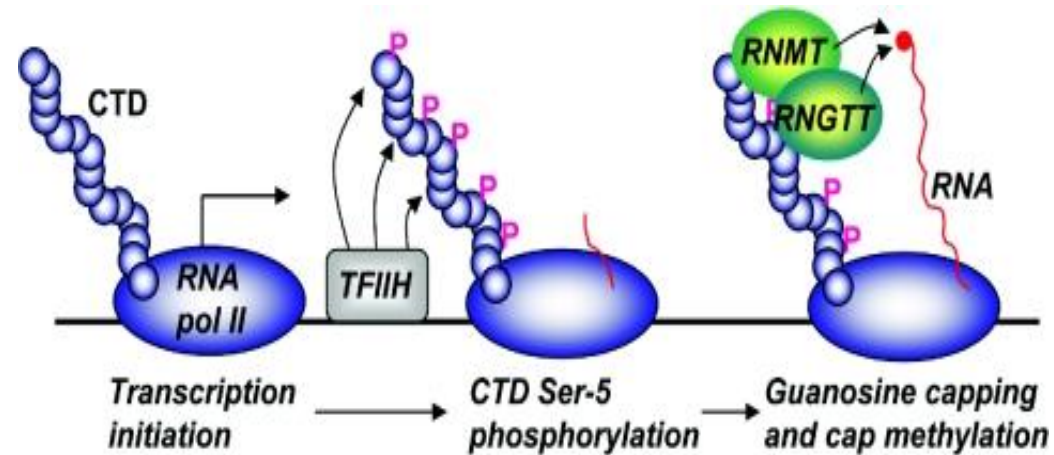
The Code of Life: mRNA Maturation-Alternative splicing

- **Alternative splicing** can give rise to **different proteins from the same gene**.
- In **most cases**, the **splicing patterns are finely regulated** so that different forms of the protein are produced at **different times** and in **different tissues**



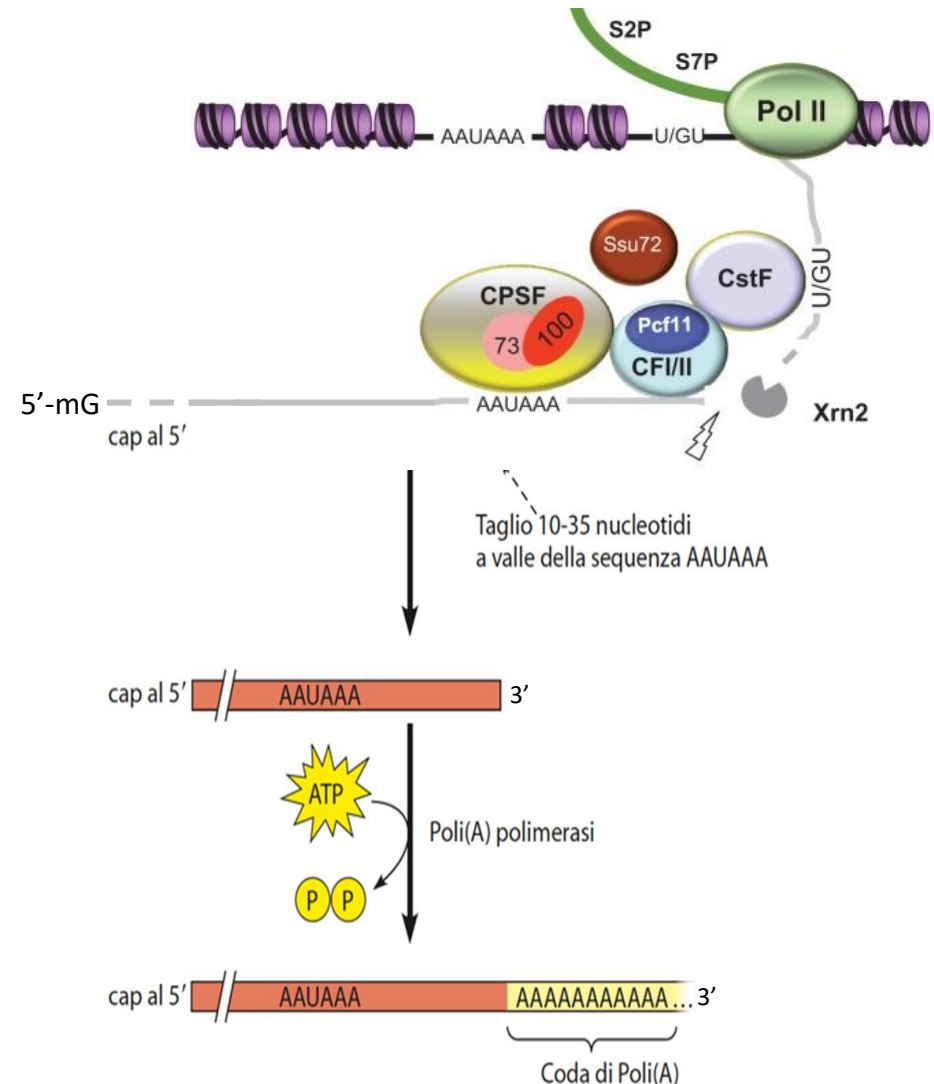
The Code of Life: mRNA Maturation-5' Capping

Almost immediately after the initiation of RNA synthesis, the **5' end of mRNA** is capped by a **methyl guanosine** residue, which **protects** it from **degradation** (by 5'-exonucleases) during elongation of the RNA chain.



The Code of Life: Transcription termination and 3' PolyAdenylation

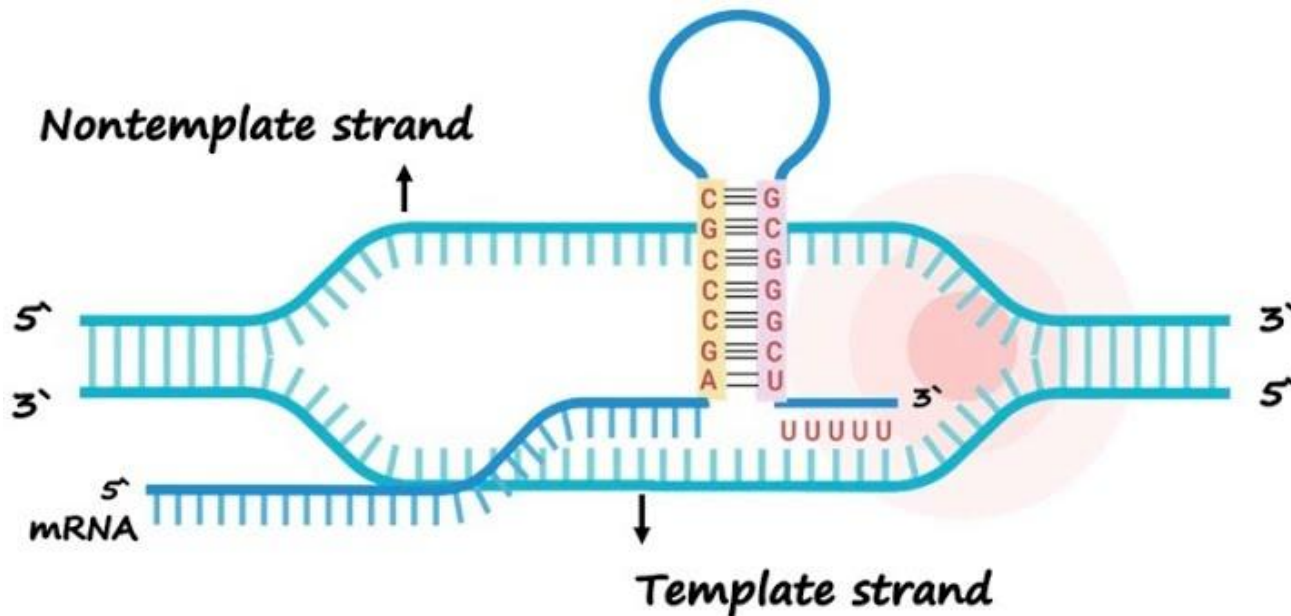
- The primary transcripts contain a highly conserved AAUAAA consensus sequence, known as a polyadenylation signal, near their 3' end.
- The polyadenylation site is recognized by a specific endonuclease that cleaves the RNA roughly 20 nucleotides downstream. Transcription may proceed for several hundred nucleotides beyond the polyadenylation site, but the 3' end of the transcript is discarded.
- The newly created 3' terminus, however, serves as a primer for addition of up to 200/250 adenine nucleotides by Poly(A) Polymerase.
- The length of poly(A) tail modulates mRNA stability protecting the mRNA from premature degradation by 3'exonucleases.



The Code of Life: Transcription Termination

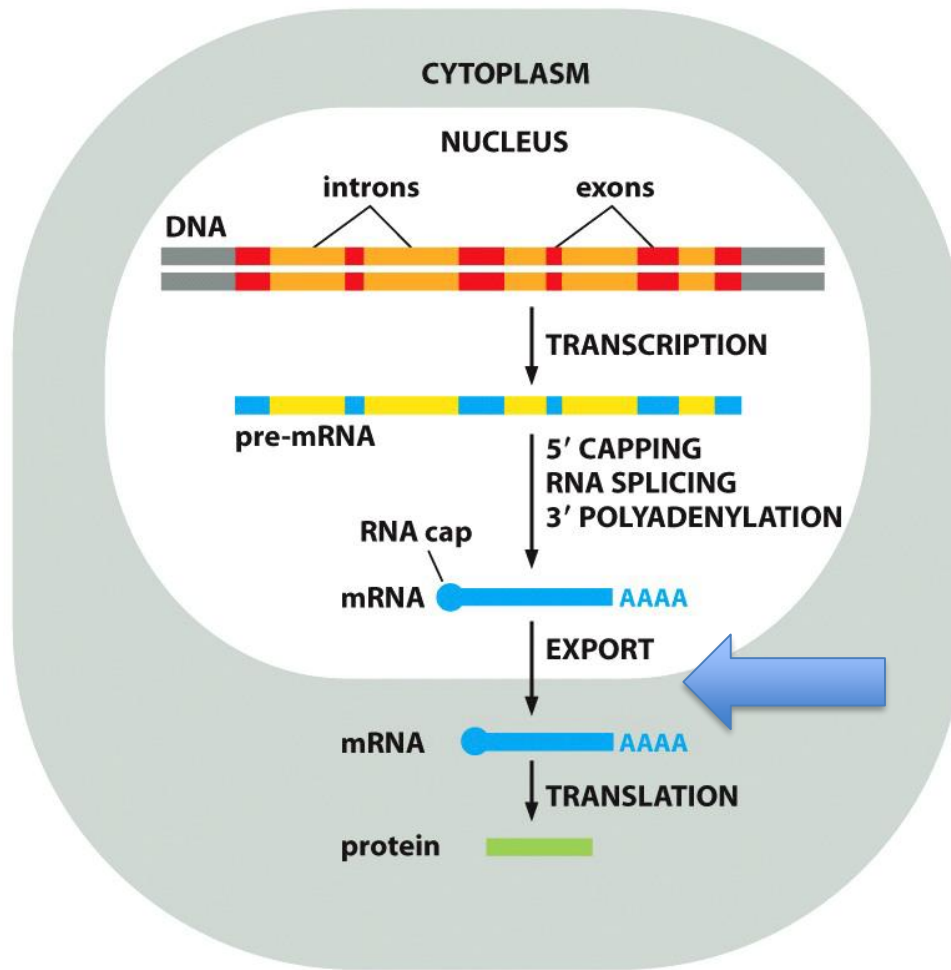
Intrinsic termination (Prokaryotes)

1. G-C rich stem loop structure is formed
2. Weak pairing of A (DNA template) : U (RNA transcript)
3. Stem-loop displaces A:U & forms more stable A (template) : T (nontemplate) base pair
4. DNA double helix "zips up" behind the RNA polymerase
5. Dissociation of RNA transcript from DNA



The Code of Life: RNA Nuclear Export

(A) EUKARYOTES



(B) PROKARYOTES

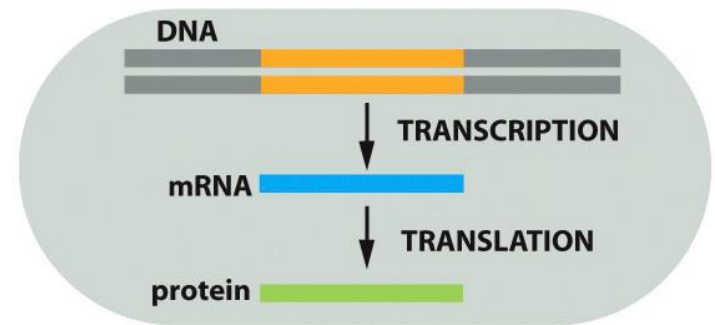
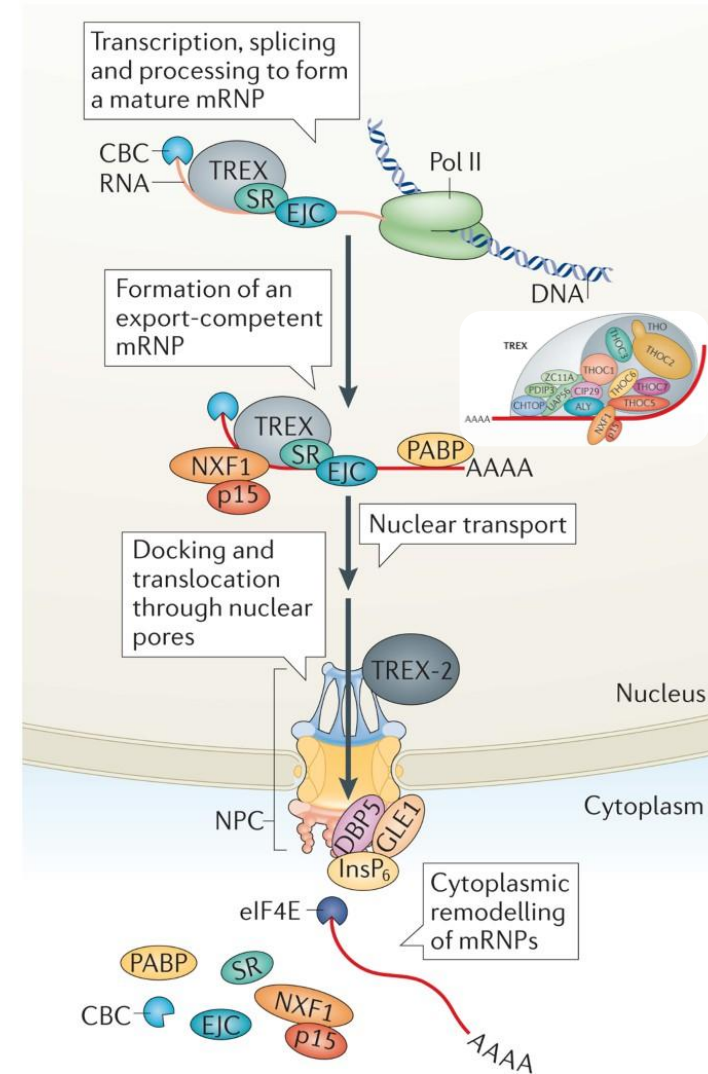


Figure 7-24 Essential Cell Biology, 4th ed. (© Garland Science 2014)

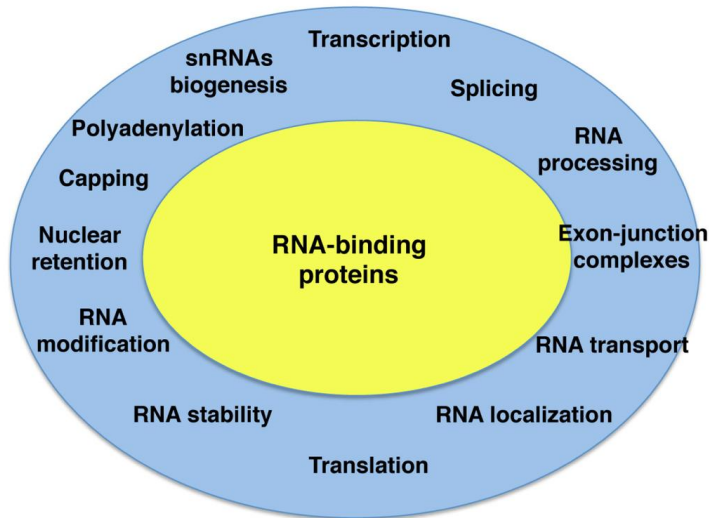
The Code of Life: Regulation of RNA Nuclear Export

Messenger Ribonucleoprotein Particles (mRNPs)

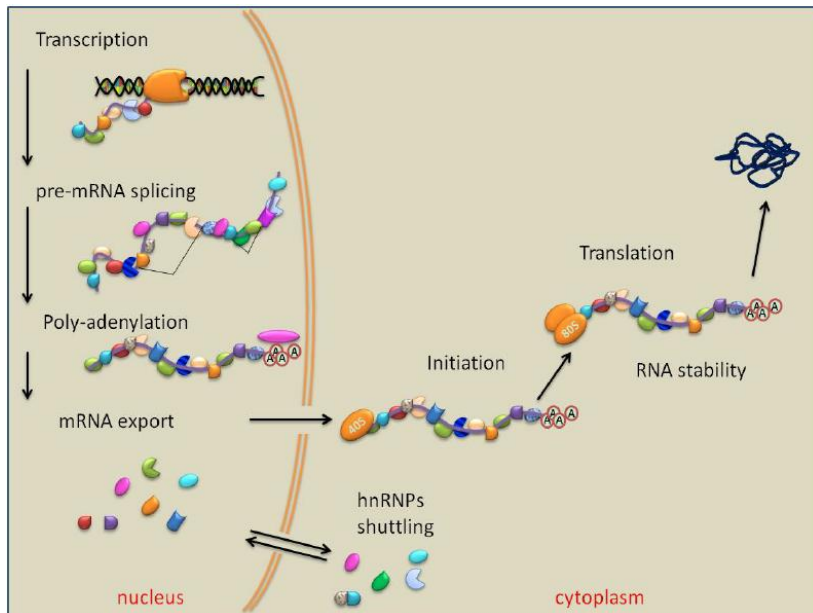
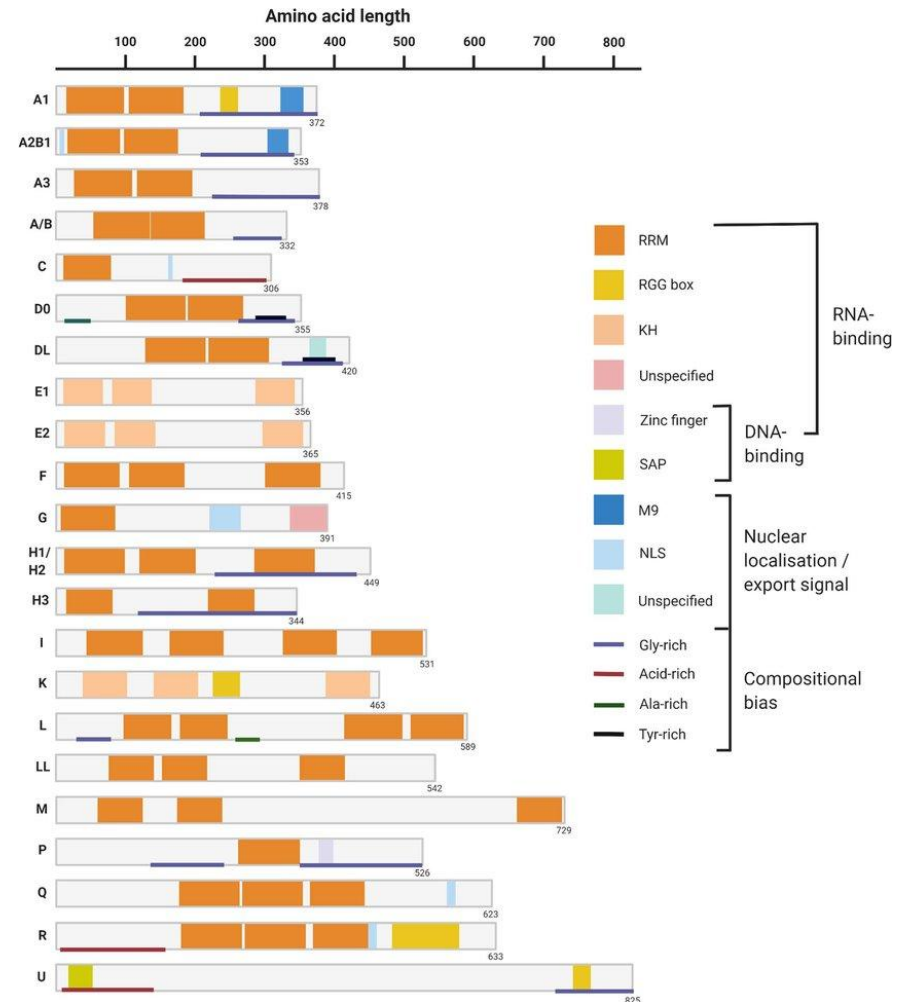
- mRNA export is an important node in regulation of mammalian gene expression.
- The cap structure (m7G) recruits the cap-binding complex (CBC) and induces RNA nuclear export.
- Two conserved complexes have key roles in the nuclear export of mRNA: transcription-export complexes TREX and TREX-2.
- Several recent examples indicate that specific biological pathways can be regulated by selective mRNA export, giving priority to some sets of mRNA over others.
- Nuclear export of functionally related sets of transcripts may be regulated by specific mRNA-binding export factors that recognize specific sequence elements among the mRNAs.



The Code of Life: Critical functions of RNA Binding Proteins



Heterogeneous nuclear ribonucleoproteins (hnRNPs)



R&D, ovvero Read & Discuss

nature reviews molecular cell biology

<https://doi.org/10.1038/s41580-022-00566-8>

Consensus statement

 Check for updates

Long non-coding RNAs: definitions, functions, challenges and recommendations

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